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Asymptotic Theory of Hamiltonian and other Systems with all Solutions Nearly Periodic*†

MARTIN KRUSKAL
Plasma Physics Laboratory, Princeton University, Princeton, New Jersey
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Consider a system of N ordinary first-order differential equations in N dependent variables, and let the independent variable s not appear explicitly. Let the system depend on a small parameter ϵ and possess a formal infinite power series expansion in ϵ, and suppose that the limiting system for $\epsilon = 0$ exists and has only periodic solutions. Then a formal solution can be constructed involving infinite power series in ϵ and satisfying the equations over large domains of s (of order $1/\epsilon$). The true solutions of the system exist over such domains and are asymptotically represented as $\epsilon \to 0$ by the formal solutions. The construction is based on the standard type of formal series solution of a "reduced" system of $N - 1$ equations in $N - 1$ dependent variables and with the new independent variable $\sigma = \epsilon s$; the omitted variable is essentially an angle variable ϕ describing the phase around the simple, closed curves. If the original system is Hamiltonian, then one can define the usual action integral $J = \int \mathbf{p} \cdot d\mathbf{q}$ to all orders; the integral is taken around the phase ring. It is proved that J is an integral of the system and that the Poisson bracket of ϕ with J is unity, both to all orders. The usefulness of this particular integral is that it is computable locally. The reduced system, after elimination of another dependent variable by means of the constancy of J, can itself be put in Hamiltonian form; if its solutions are nearly periodic, the whole procedure can be reapplied. The present theory encompasses previous proofs of adiabatic invariance to all orders for particular systems such as the harmonic oscillator, the nonlinear oscillator, the charged particle gyrating tightly in a given electromagnetic field, and the longitudinal back-and-forth motion of such a particle trapped between two "magnetic mirrors" in a weak electric field. There are many other applications.

A. INTRODUCTION

1. Historical Perspective

THE phenomenon of adiabatic invariance (constancy of an action integral under slow change of external parameters) has long been known in classical mechanics, receiving prominence because of its role in the so-called quantum conditions and of their importance during the transition from classical to quantum mechanics. Although the invariance was recognized to be only approximate and was only demonstrated to the lowest significant order,[1] the question of whether the constancy might be valid to higher order seems not to have been considered at that time or in that connection. For this there were perhaps two reasons: the first that the question probably appeared to be of only academic interest, and the second that not all adiabatic invariants are constant beyond lowest order.

The question apparently first became of significance for applications in connection with the magnetic moment of gyration of a charged particle gyrating tightly in a strong magnetic field, which was shown to be an adiabatic invariant by Alfvén.[2] On the one hand, the theory of virtually every prospective device for the production of useful energy from controlled thermonuclear fusion[3] has leaned very heavily on the constancy of this magnetic moment, and in those cases for which more or less steady operation was envisioned (stellarator, mirror machine, etc.) it was seen that the requirement that particles remain confined for periods of time encompassing many millions of gyrations could generally be met only if the magnetic moment were in fact constant to a much higher approximation. In connection with the stellarator several other related approximate results were proved to hold to all orders,[4,5] which made it seem both fitting and somewhat more likely that the constancy of the magnetic moment would follow suit. On the other hand, astrophysical theories (e.g. the Fermi mechanism[6] for cosmic-ray production) led to precisely the same question, in response to which

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Hellwig[7] proved the constancy of the magnetic moment to the next order beyond the lowest. Kulsrud[8] considered the simpler problem of a harmonic oscillator with slowly changing coefficient of elasticity and proved that its adiabatic invariant (ratio of energy to frequency) was constant to all orders, seemingly the first result of the kind. In quick succession after this breakthrough Kruskal[9] proved the analogous result for the gyrating particle and Lenard[10] for the anharmonic oscillator. Berkowitz and Gardner[11] stiffened the results with unwonted (but wanted!) mathematical rigor by proving that the formal expansions employed to describe the motion of the gyrating particle really were correct asymptotic series for the true trajectory.

Another adiabatic invariant whose degree of constancy has become of practical importance recently is the so-called longitudinal adiabatic invariant of the gyrating particle. This obtains under more restrictive conditions than the magnetic moment invariant does, namely, if the tightly gyrating particle moves back and forth along a magnetic line of force and periodically returns nearly to its initial state. It seems to have been first employed (to lowest order) by Rosenbluth.[12] It became of interest again (and to higher order) in connection with the Van Allen belt.[13] Gardner[14] has settled the matter by proving the invariance to all orders.

2. Significance and Arrangement of Present Paper

In the present paper, a unification and simplification of all these preceding treatments of adiabatic invariance to higher order is achieved. The significant common element has turned out to be that each was concerned with a Hamiltonian system whose solutions are all nearly periodic. Lest this appear too like a truism, it may be pointed out that there are other types of adiabatic invariant associated with Hamiltonian systems whose solutions to lowest order are ergodic over surfaces of constant energy in phase space. (Incidentally, Lenard[10] has shown that descriptions and results valid to all orders can be obtained even for a system all of whose solutions are nearly only "multiply periodic" so long as it is linear.)

These two characteristics, of being Hamiltonian and of having all solutions nearly periodic, may largely be treated independently of each other, although their amalgamation (coined "almost-mechanics" by E. Gerjuoy) leads to interesting consequences (part E). Since the theory of Hamiltonian systems is already extremely well developed, everything we need is at hand, except, perhaps, for the result proved in Appendix 2. We therefore devote the major portion of the paper to discussing systems of differential equations with all solutions nearly periodic, which is of great independent interest in any case. We first derive the appropriate formal series solutions (Secs. B. 5–9). This treatment is in essence very similar to the method of Kryloff and Bogoliuboff,[15] but seems to be a generalization inasmuch as the latter method is confined to quasi-linear second-order systems.

Following the determination of the formal series we prove (Secs. B. 10–11) that the given system has exact solutions for an appropriately large range of the independent variable and that the formal series solutions represent the exact solutions asymptotically. The method seems simpler and more natural than that given by Berkowitz and Gardner[11] for the special case of the gyrating particle. An interesting feature is the "bootstrap" argument (Sec. B. 11) by which the two items are linked together and proved simultaneously, thereby avoiding a certain duplication to be found in their treatment.

Part C starts with the theorem of phase independence, the importance and utility of which for this paper can hardly be overestimated: it is fair to say that parts C, D, and E consist of almost nothing but its systematic exploitation. The full power of this ridiculously simple theorem can probably only be appreciated by those who, like the author, have spent tremendous amounts of effort and time in attempting (often unsuccessfully) to obtain various results to all orders.

3. Notation and Conventions

We consistently employ vector and dyadic notation in dealing with arrays of quantities and often indulge in the geometric terminology which accompanies it so naturally. Vectors are boldfaced throughout, and polyadics are represented by script letters. The number of components (array elements)

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[13] J. Van Allen, J. Geophys. Research **64**, 1683 (1959).
[14] C. Gardner, Phys. Rev. **115**, 791 (1959).
[15] N. Kryloff and N. Bogoliuboff, *Introduction to Non-Linear Mechanics*, (Annals of Mathematics Studies, No. 11, Princeton University Press, Princeton, 1947).

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of every vector (etc.) will of course be announced (if it is definite) when that vector is first introduced, but that number will not be indicated, thereafter, by any special notational feature. The unit dyadic in any number of dimensions will be denoted by $\mathcal{I}$. The inner ("dot") product of two vectors is defined only if they have the same number of components, and the double inner product is to be understood in the sense that $\mathbf{ab} : \mathbf{cd} = \mathbf{a}\cdot\mathbf{d}\,\mathbf{b}\cdot\mathbf{c}$.

Differentiation, whether ordinary or partial, will be consistently denoted by attaching as a subscript the variable with respect to which the derivative is taken. Moreover, the use of subscripts is reserved exclusively for this purpose. To maintain the correct vector ordering, it is convenient, in a few places, to put the subscript before the symbol to which it is attached; thus $_{\mathbf{x}}\mathbf{f}$ denotes the gradient of the vector field $\mathbf{f}$ and the transpose of the dyadic $\mathbf{f}_{\mathbf{x}}$.

When one set of variables (coordinates) is transformed into another, which will be symbolized by a double-headed arrow $\leftrightarrow$, the new variables will be denoted by symbols entirely distinct from the old ones. Therefore (in contrast, for instance, with the conventional notation in thermodynamics), there can arise no possible ambiguity as to what is being held fixed during a partial differentiation.

We will have mostly to do with formal infinite series in increasing integral powers of an expansion parameter ϵ, and we generally omit any indication of ϵ in the notation. If $\mathbf{f}$ denotes such a series, $\mathbf{f}^{(n)}$ denotes the coefficient of ϵ^n, and $\mathbf{f}^{[n]}$ denotes the sum of all the terms of $\mathbf{f}$ up to and including $\epsilon^n\mathbf{f}^{(n)}$. As usual, $O(\epsilon^n)$ denotes any quantity which, as $\epsilon \to 0$, approaches zero at least as fast as ϵ^n.

A variable will be called *angle-like* if it is determined up to and only up to an additive integer, so that it is like an angle variable measured in units of a complete revolution. (As here, defined terms will regularly be italicized when they are introduced.)

4. Local Dependence and Useful Integrals

For a full exposition of the significance of the results to come, it is necessary to discuss here the concept of local dependence. A function $\eta(\mathbf{x})$ on a space of points $\mathbf{x}$ which is a functional of (i.e. depends on) another such function $\xi(\mathbf{x})$ depends *locally* on $\xi(\mathbf{x})$ if the value of η at any given point $\mathbf{x}$ is independent of the values of ξ outside of arbitrarily small neighborhoods of that point. (Here ξ and η can stand indifferently also for sets of independently many functions.) In practice this usually means that the value of η at $\mathbf{x}$ is a function of $\mathbf{x}$ and of the values at $\mathbf{x}$ of ξ and its derivatives (and often in fact only a finite number of them). We shall use this definition only "metamathematically" and so may modify it or use it loosely on occasion.

To see why local dependence is significant, let us observe that one of the later results is the derivation of an approximate integral or "constant of motion" J (Sec. E. 2) for certain systems of differential equations. How can this be important, in view of the fact that such systems commonly have not merely one but a complete set of exact integrals? The answer is that J is a "useful" integral. There are two ways in which an integral of a system may fail to be useful. The first is that it may fail to be a so-called isolating integral (see Wintner[16]), the second that it may not be locally computable.

To illustrate the first possibility (without really going into the subject or even giving the definition of isolating integral) by means of a really trivial example, let us take the system which consists of two independent subsystems

$$\alpha_s = a, \qquad \beta_s = b, \tag{A1}$$

where α and β are angle-like variables, s is the independent variable, and a and b are given constants. The subsystems have each one integral, namely,

$$\alpha - as, \qquad \beta - bs, \tag{A2}$$

respectively. If we now ask for integrals which are independent of s, we see that the subsystems have none, but the combined system has the s-independent integral

$$b\alpha - a\beta. \tag{A3}$$

This integral, however, is multivalued. If a and b are commensurable (a/b rational) it is an isolating integral and is useful: Knowing the value of the integral (from initial conditions, for example) and, say, of β, we can say something about the value of α; in fact, writing a/b in standard form as m/n (m and n relatively prime integers and n positive), we can tell that α must have one of n (essentially distinct) values all of whose pairwise differences are multiples of $1/n$. On the other hand, if a and b are incommensurable, the integral is not isolating, and is clearly not useful in the same way.

The second possibility is of more direct concern. For illustration consider the motion of a particle in a given static three-dimensional potential field. To guarantee the existence of a complete set of isolating integrals let us agree to accept time-dependent integrals. (In various special cases this

[16] A. Wintner, *Analytical Foundations of Celestial Mechanics* (Princeton University Press, Princeton, 1941) Sec. 128.

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can be guaranteed even for time-independent integrals, for instance, if the potential has a translational invariance, so that the corresponding spatial coordinate can be ignored, grows linearly in time, and can serve in effect as the time.) In such a case, it is sometimes said that the energy is the only integral. Strictly speaking this is nonsense, since we know that there is a complete set of integrals (for instance, the values of the dynamical variables taken on at $t = 0$, viewed as functions of the free dynamical variables). But there is no question about the exceptional status of the energy integral: In countless physical arguments it plays a unique role denied to the other integrals. The reason for this is that the energy depends only locally on the potential and there is no other integral (functionally independent of the energy) with that property. The point is that, although in every particular case (i.e., for every particular potential function) the other integrals exist, and can be perfectly well defined mathematically, the problem of finding them is equivalent to the problem of integrating the equations of motion of the particle. Only the energy integral is useful because it alone is known *a priori*, without integrating the system completely, and because it alone remains unaffected (in its functional form) in one region by a change in the potential elsewhere along the orbit of the particle.

5. Autonomous Systems

Throughout this paper we shall be dealing with systems of ordinary differential equations. Primarily for notational convenience we do not permit the functions expressing the derivatives to depend on the independent variable, but this is no essential restriction, for there is no objection to such a dependence *de facto* (so long as appearances are preserved *de nomine*) by way of the existence of a dependent variable which always equals the independent variable, i.e., has derivative unity. Denoting the dependent variables collectively by $\mathbf{x}$ and the independent variable by s, we may write a general system in the form

$$\mathbf{x}_s = \mathbf{f}(\mathbf{x}). \tag{A4}$$

This may be interpreted as a first-order differential equation in the space of points $\mathbf{x}$. We call it an *autonomous* system for short and as a faint reminder that the rate of change of $\mathbf{x}$ depends only on $\mathbf{x}$ itself. When $\mathbf{f}(\mathbf{x})$ is a formal series in powers of ϵ, we shall say that (A4) is in *standard form* if $\mathbf{f}$ starts with a term of zeroth order which nowhere (as a function of $\mathbf{x}$) vanishes (in all components simultaneously).

We assume that $\mathbf{f}$ is as smooth as needed, so all the standard theory of systems of ordinary differential equations applies. A most fundamental teaching of that theory is that for every specified "initial condition," i.e., specification of $\mathbf{x}$ at (say) $s = 0$, there exists a solution $\mathbf{x}(s)$ of (A4) which may be continued indefinitely so long as $\mathbf{x}$ stays within the domain of definition of $\mathbf{f}$. Another fundamental teaching is that such a solution is unique: Any two functions satisfying the same autonomous system and equal for $s = 0$ are equal for all values of s (in any interval which contains zero and over which both functions are defined); this is in fact a special case of the fundamental approximation theorem for ordinary differential equations (Appendix 1).

Let $\mathbf{X}(\mathbf{x}, s)$ denote the unique solution of (A4) which passes through $\mathbf{x}$ at $s = 0$, i.e., let $\mathbf{X}$ be defined by the conditions

$$\mathbf{X}_s(\mathbf{x}, s) = \mathbf{f}(\mathbf{X}(\mathbf{x}, s)), \tag{A5}$$

$$\mathbf{X}(\mathbf{x}, 0) = \mathbf{x}. \tag{A6}$$

An important property of the solutions of autonomous systems is that the one-parameter family of mappings of the form $\mathbf{x} \to \mathbf{X}(\mathbf{x}, s)$ (between parts of $\mathbf{x}$ space) constitute a group, and furthermore that the group operation on $\mathbf{X}(\mathbf{x}, s')$ and $\mathbf{X}(\mathbf{x}, s)$ is $\mathbf{X}(\mathbf{x}, s' + s)$, so that the group is isomorphic to the additive group of real numbers s. [Indeed, (A6) already states that $s = 0$ represents the identity.] To show this one must establish that

$$\mathbf{X}(\mathbf{X}(\mathbf{x}, s'), s) = \mathbf{X}(\mathbf{x}, s + s'). \tag{A7}$$

This follows from the uniqueness theorem, since the two sides are equal for $s = 0$ by (A6), and satisfy the same autonomous system because the left-hand side is obtained from $\mathbf{X}(\mathbf{x}, s)$ merely by substituting $\mathbf{X}(\mathbf{x}, s')$ for $\mathbf{x}$ and the right-hand side by substituting $s + s'$ for s, neither of which substitutions invalidates (A5).

A further property we use in the next section is now easily obtained. Using (A5) with $s = 0$ and (A6) we have

$$\mathbf{X}_s(\mathbf{x}, 0) = \mathbf{f}(\mathbf{x}). \tag{A8}$$

Differentiating (A7) with respect to s', setting $s' = 0$, and using (A6) and (A8) gives

$$\mathbf{X}_{\mathbf{x}}(\mathbf{x}, s) \cdot \mathbf{f}(\mathbf{x}) = \mathbf{X}_s(\mathbf{x}, s). \tag{A9}$$

6. Recurrent Systems

If all solutions of (A4) are periodic (not necessarily with equal periods) we call (A4) a *recurrent*

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system. We also call the vector field $\mathbf{f}$ in $\mathbf{x}$ space *recurrent* under the same circumstances, namely, when its integral curves (curves which at each of their points are tangent to the field) are closed, forming topological circles; we call them *loops*. For each $\mathbf{x}$ there is then a least positive value of s for which the initial value $\mathbf{x}$ recurs; we denote this period by $S(\mathbf{x})$ and have

$$\mathbf{X}(\mathbf{x}, S(\mathbf{x})) = \mathbf{x}, \tag{A10}$$

$$\mathbf{X}(\mathbf{x}, s) \neq \mathbf{x} \quad \text{for} \quad 0 < s < S(\mathbf{x}). \tag{A11}$$

We next show by a formal argument the obvious fact that the period $S(\mathbf{x})$ is constant along a solution of (A4). We first take the s derivative of (A10) as $\mathbf{x}$ varies in accordance with (A4), or in other words, we take the $\mathbf{x}$ derivative, on the right, say, and then dot on the right with $\mathbf{f}(\mathbf{x})$; using (A9) with $s = S(\mathbf{x})$ the result may be written

$$\mathbf{X}_s(\mathbf{x}, S(\mathbf{x})) + \mathbf{X}_s(\mathbf{x}, S(\mathbf{x}))S_{\mathbf{x}}(\mathbf{x})\cdot\mathbf{f}(\mathbf{x}) = \mathbf{f}(\mathbf{x}). \tag{A12}$$

But setting $s = S(\mathbf{x})$ in (A5) we have

$$\mathbf{X}_s(\mathbf{x}, S(\mathbf{x})) = \mathbf{f}(\mathbf{x}) \tag{A13}$$

in view of (A10). Therefore, the extreme terms in (A12) cancel, and, furthermore, since $\mathbf{f} \neq 0$, we have

$$S_{\mathbf{x}}(\mathbf{x})\cdot\mathbf{f}(\mathbf{x}) = 0, \tag{A14}$$

which is the desired result; to be entirely explicit,

$$[S(\mathbf{X}(\mathbf{x}, s))]_s = S_{\mathbf{x}}(\mathbf{X}(\mathbf{x}, s))\cdot\mathbf{X}_s(\mathbf{x}, s)$$
$$= S_{\mathbf{x}}(\mathbf{X}(\mathbf{x}, s))\cdot\mathbf{f}(\mathbf{X}(\mathbf{x}, s)) = 0 \tag{A15}$$

in view of (A5) and of (A14) with $\mathbf{X}(\mathbf{x}, s)$ for $\mathbf{x}$, so that the period evaluated at all points of the solution $\mathbf{X}(\mathbf{x}, s)$ is the same (independent of s). This result may be written

$$S(\mathbf{X}(\mathbf{x}, s)) = S(\mathbf{x}). \tag{A16}$$

Another hardly surprising result, a generalization of (A10), is

$$\mathbf{X}(\mathbf{x}, s + S(\mathbf{x})) = \mathbf{X}(\mathbf{x}, s). \tag{A17}$$

This can be deduced from the uniqueness theorem, or by taking (A7) with $s' = S(\mathbf{x})$ and using (A10). It is now obvious that $\mathbf{X}(\mathbf{x}, s)$ has the same value for two different values of s if and only if they differ by an integral multiple of $S(\mathbf{x})$; the "if" follows by induction from (A17), and the "only if" part since otherwise a contradiction to (A11) could be similarly obtained.

7. Splittable Systems

An important property of some systems of differential equations is the possession of an autonomous subsystem. Suppose that, of the set of N dependent variables denoted collectively by $\mathbf{x}$, there are some M (where $1 \leq M \leq N - 1$) whose s derivatives as given by (A4) are independent of the remaining $N - M$ variables. The M variables then satisfy an autonomous subsystem of their own and we call the original system *split*. More generally, we call a system *splittable* if by a local change of dependent variables it can be transformed into a split system, i.e., if there are local definitions of M new variables, functions of $\mathbf{x}$ whose derivatives, computed using (A4), are expressible as functions of the M new variables themselves alone. These variables then satisfy a *new autonomous* system of fewer variables.

By a local change of variables we mean, a change whose defining formulas depend locally on $\mathbf{f}(\mathbf{x})$ (see Sec. 4). The requirement of localness is essential to the definition; without it, for instance, every system with a complete set of isolating integrals would be not merely splittable but completely splittable into N independent autonomous subsystems of one dependent variable each.

The problem of solving a system of order N may, if the system is splittable, be split up into the problems of first solving a system of order M and then one of order $N - M$.

B. NEARLY RECURRENT SYSTEMS

1. Formulation and Description

Consider the system of first-order ordinary differential equations

$$\mathbf{x}_s = \mathbf{F}(\mathbf{x}, \epsilon), \tag{B1}$$

where $\mathbf{x}$ and $\mathbf{F}$ are vectors of N components. We suppose that $\mathbf{F}$ is defined in a suitable domain of its arguments, with properties to be described shortly, and that it and all its derivatives of all orders exist and are continuous. Let the domain be the product of a domain in $\mathbf{x}$ space with a closed interval of values for ϵ among which occurs zero. Let (B1) be a recurrent system for $\epsilon = 0$, so that $\mathbf{F}(\mathbf{x}, 0)$ is a nowhere vanishing vector field with all integral curves closed. [When we speak of loops in connection with (B1), we shall always mean these, even if it should happen, most atypically, that (B1) is also recurrent for $\epsilon \neq 0$.] We require finally that the domain of points in $\mathbf{x}$ space for which $\mathbf{F}(\mathbf{x}, \epsilon)$ is defined be closed, bounded, and N dimensional and be made up of loops.

As ϵ approaches zero any solution of (B1) gets closer and closer to a loop, so long as the range of

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values of s considered remains bounded (independently of ϵ). We may say that the solutions of system (B1) are all *nearly periodic*[17] for small ϵ. The distance in $\mathbf{x}$ space (in any reasonable sense) by which a solution misses its initial point after one *gyration*, namely, after approximately following one closed loop around, is of order ϵ; in general, it is to be expected that these small deviations will accumulate, so that after a large number of gyrations (a large change in s) of order $1/\epsilon$, the solution will have *drifted* a finite amount and will be gyrating in a different region of $\mathbf{x}$ space from where it started. The curve in $\mathbf{x}$ space traversed by a solution in its entirety may be suggestively thought of as the distorted image of a helix of small pitch, as shown in Fig. 1.

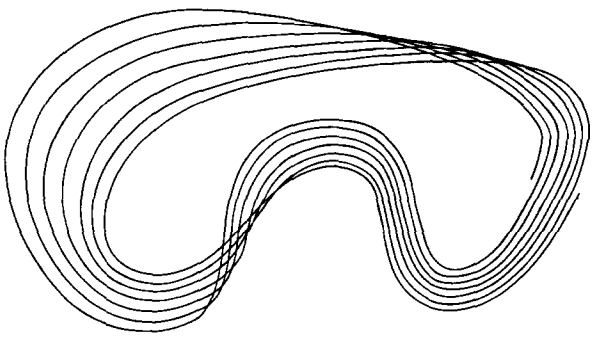


FIG. 1. Curve traversed by a solution of (B1). The curve cannot cross itself; the apparent crossings are due to projection.

2. Inadequacy of Obvious Series Solution

Our first task is to find some kind of formal solution of (B1), in terms of infinite power series in ϵ, which is asymptotically valid uniformly over large ranges of s and shows explicitly its nearly periodic nature. The obvious approach at first sight seems to be to represent $\mathbf{x}$ directly by a series,

$$\mathbf{x} \approx \mathbf{u}^{(0)}(s) + \epsilon\,\mathbf{u}^{(1)}(s) + \epsilon^2\,\mathbf{u}^{(2)}(s) + \cdots, \tag{B2}$$

and to determine the $\mathbf{u}^{(n)}$ by substituting this series into (B1), expanding the function on the right-hand side in a formal double Taylor series around its zeroth-order arguments $(\mathbf{u}^{(0)}, 0)$, and collecting and separately equating like powers of ϵ. This leads to the equations

$$\mathbf{u}_s^{(0)} = \mathbf{F}(\mathbf{u}^{(0)}, 0),$$
$$\mathbf{u}_s^{(1)} = \mathbf{F}_x(\mathbf{u}^{(0)}, 0)\cdot\mathbf{u}^{(1)} + \mathbf{F}_\epsilon(\mathbf{u}^{(0)}, 0), \cdots, \tag{B3}$$

which do indeed determine the $\mathbf{u}^{(n)}$ if some initial values are given, say at $s = 0$. [And, in fact, if $\mathbf{F}$ is analytic then the series in (B2) converges to an analytic solution of (B1) for sufficiently small ϵ.] Of course, $\mathbf{u}^{(0)}$ runs periodically around and around a loop. But (B2) obviously can represent a solution asymptotically only so long as the solution stays within $O(\epsilon)$ of $\mathbf{u}^{(0)}$, which in general means only for bounded values of s: For larger s the solution will have drifted farther from its initial loop. Therefore (B2) is not useful for our purpose.

3. Standardization

Before proceeding any further let us replace (B1) by

$$\mathbf{x}_s = \mathbf{f}(\mathbf{x}), \tag{B4}$$

where $\mathbf{f}(\mathbf{x})$ denotes the Taylor expansion

$$\mathbf{f}(\mathbf{x}) = \sum_{n=0}^{\infty} \frac{\epsilon^n}{n!}\left[\frac{d^n\mathbf{F}}{d\epsilon^n}\right]_{\epsilon=0} \tag{B5}$$

of $\mathbf{F}$ around $\epsilon = 0$, and is therefore an infinitely differentiable formal infinite series of non-negative powers of ϵ. We have two reasons for doing this: First, since we are seeking a formal solution (which later is proved to represent the true solution asymptotically, but which does not in general converge to it, nor indeed at all; see the next section) we must switch over to formal series at some point anyway, and it is simplest to do so at the start. Second, the theory about to be developed can be applied not only to true systems of differential equations, but also to merely formal systems (and in particular to the "reduced" system resulting from the first application of the theory; see Secs. C.3 and E.6).

However, a warning is in order. Certainly (B4) is the formal analog of (B1); and indeed, the formal solutions of (B4) which we shall obtain, when taken up to terms of order n, do provide approximations up to that order of the true solutions of (B1) [as do also the true solutions of the system $\mathbf{x}_s = \mathbf{f}^{[n]}(\mathbf{x})$], as we see from the basic approximation lemma of Appendix 1. But this is only for ranges of s bounded independently of ϵ; due to the exponential in the estimate (F9), no matter how large we take n, we could not expect these formal approximations to remain at all close to the true solutions for ranges of s of order $1/\epsilon$ (or of order $1/\epsilon^\alpha$, $\alpha > 0$; but we could, for instance, for ranges of order $\log \epsilon$). Or rather, we could not expect so except for the near periodicity assumed; one of our conclusions (Sec.

[17] This concept is not to be confounded with that of almost periodic functions, H. Bohr, *Fastperiodische Funktionen, Ergebnisse der Mathematik und Ihrer Grenzgebiete* (Julius Springer, Berlin, 1932), B.1, N.5, which we are not concerned with here.

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B.10) is that the approximations are in fact good for ranges of order $1/\epsilon$. (But not quite so good: two orders in ϵ get lost in the shuffle.)

4. An Example

An extremely simple but doubly illuminating example is provided by the harmonic oscillator with slowly varying rest point. As our basic equation we take

$$\epsilon^2 u_{tt} + u = r(t), \tag{B6}$$

where u is the displacement of the oscillator and $r(t)$ the rest point, a given function of its argument (and allowably a series in powers of ϵ). If r were constant, every solution of (B6) would oscillate sinusoidally around r with period $2\pi\epsilon$; as it is, r is nearly constant during a period, so it should be possible to put (B6) into the form this paper deals with.

To render the period finite we introduce $s \equiv t/\epsilon$, so (B6) becomes

$$u_{ss} + u = r(\epsilon s), \tag{B7}$$

a version of the equation we might have started with; here the right-hand side varies slowly because of the factor ϵ in the argument. To render (B7) as a formally first-order system we introduce $v \equiv u_s$. To eliminate the dependence (of r) on the independent variable s we bring back t, but now as a new dependent variable. (It wouldn't do to introduce a new dependent variable equal to s itself, because s does not recur, even to lowest order.) Altogether, we end up with a system in standard form (B4) with

$$\begin{aligned} \mathbf{x} &\equiv (u, v, t), \\ \mathbf{f} &\equiv (v, r(t) - u, \epsilon). \end{aligned} \tag{B8}$$

This system is nearly recurrent because to zeroth order, t remains constant and u and v oscillate periodically.

Having used this example to illustrate how a given system may be transformed as required by the theory, we now use it to illustrate why the formal series we are led to are, in general, only asymptotic, and not convergent as one might have hoped (when $\mathbf{f}$ is convergent, i.e., analytic). For this we need only the original form (B6). Since the corresponding homogeneous equation has the general solution

$$u = \alpha \cos(t/\epsilon) + \beta \sin(t/\epsilon), \tag{B9}$$

we may confine ourselves to a particular solution of (B6). One particular solution, for that matter, may be singled out as "oscillation free" (a concept which may be rendered precise by interpreting it to mean the vanishing of J; see Sec. E.2). This is a unique characterization to all orders, though ill defined for finite ϵ. The oscillations referred to are naturally those of small period $2\pi\epsilon$, and are excluded by requiring that t differentiation not affect (not lower) the order in ϵ, in obvious contrast to its effect on the general solution (B9). Thus $\epsilon^2 u_{tt}$ is regarded as small, and (B6) is recursively solvable to yield

$$\begin{aligned} u &= r - \epsilon^2 u_{tt} \\ &= r - \epsilon^2 r_{tt} + \epsilon^4 r_{tttt} - \cdots . \end{aligned} \tag{B10}$$

This formal development differs from the Taylor series for $r(t + \epsilon)$ in two respects unimportant for present purposes (the absence of odd powers of ϵ and the alternation of sign among the even powers) and in one important respect, the absence of $n!$ in the denominator of the ϵ^n term. It is thus clear that even the analyticity of $r(t)$ is insufficient to assure the convergence of (B10). For (B10) to have a radius of convergence greater than zero, the development of $r(t + \epsilon)$ must have an infinite radius of convergence, i.e., $r(t)$ must be an entire function. And even this is not quite enough, as can be seen by Fourier analysis of (B6).

5. More Appropriate Variables

Let us start our search for an asymptotic solution of (B4) by introducing a new coordinate system more appropriate than $\mathbf{x}$ for describing a nearly periodic but slowly drifting trajectory. To this end observe that the loops form an $(N-1)$-dimensional family, so that there exist $N-1$ independent functions of $\mathbf{x}$ each of which is constant on every loop; these can obviously also be chosen infinitely differentiable, since $\mathbf{f}^{(0)}$ is. [For instance, we may pick an infinitely differentiable $(N-1)$-dimensional hypersurface in $\mathbf{x}$ space which is nowhere tangent to $\mathbf{f}^{(0)}$, then pick any set of $N-1$ independent infinitely differentiable coordinates $\mathbf{Y}$ on the hypersurface, and then extend $\mathbf{Y}$ to the rest of $\mathbf{x}$ space so as to be constant on every loop.] Denote some arbitrary fixed set of such functions collectively by a vector function $\mathbf{Y}(\mathbf{x})$ with $N-1$ components. Thus each loop is characterized uniquely by the constant (vector) value assumed on it by $\mathbf{Y}(\mathbf{x})$.

Next, let $\Upsilon(\mathbf{x})$ be any infinitely differentiable angle-like function which varies monotonically around each loop; for definiteness say it increases with s. We then change variables from $\mathbf{x}$ to $\mathbf{y}$ and υ

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by the transformation

$$\mathbf{y} = \mathbf{Y}(\mathbf{x}), \qquad \upsilon = \Upsilon(\mathbf{x}). \tag{B11}$$

Of course, υ is determined only up to an arbitrary additive integer. The transformation clearly possesses an infinitely differentiable inverse which we denote by

$$\mathbf{x} = \mathbf{X}(\mathbf{y}, \upsilon), \tag{B12}$$

where $\mathbf{X}$ is, of course, periodic in υ (with period unity).

In terms of our new variables the system (B4) takes the form

$$\mathbf{y}_s = \epsilon\, \mathbf{g}(\mathbf{y}, \upsilon), \qquad \upsilon_s = \psi(\mathbf{y}, \upsilon), \tag{B13}$$

where (omitting the arguments $\mathbf{y}$, υ consistently)

$$\epsilon\mathbf{g} \equiv \mathbf{Y}_{\mathbf{x}}(\mathbf{X})\cdot\mathbf{f}(\mathbf{X}), \tag{B14}$$

$$\psi \equiv \Upsilon_{\mathbf{x}}(\mathbf{X})\cdot\mathbf{f}(\mathbf{X}); \tag{B15}$$

here $\mathbf{g}$ and ψ are infinitely differentiable formal series, their leading terms are $O(1)$ and that of ψ is definitely positive, and $\mathbf{g}$ is written accompanied by the factor ϵ because, by construction, $\mathbf{y}_s$ must vanish to zeroth order. These equations exhibit the gyration clearly: to zeroth order, as s increases $\mathbf{x}$ stays on a loop (since y is constant) and runs periodically around it (since υ increases monotonically); to first order the loop may drift (since $\mathbf{y}$ may vary).

We can hardly rest content with our system in the form (B13), however. For, the gyration it depicts may be very uneven, full of sudden accelerations and decelerations, since ψ may fluctuate wildly; further, and, in fact, more serious, the net rate of drift is not at all in evidence, since in the course of one gyration $\mathbf{g}$ may point in all different directions, even quite opposite to the direction that $\mathbf{y}$ is effectively drifting in, which can only be obtained by some sort of integration around a loop. Using a previous metaphor, we might say that we have introduced coordinates $\mathbf{y}$, υ in terms of which every trajectory does indeed look like a helix of small pitch (if we think of υ as the angle of a point on a circle), but only to zeroth order, and even there s does not increase uniformly around the loops.

But these defects can be remedied. As for one minor issue, it is not hard to see that if Υ had been chosen judiciously, $\psi^{(0)}$ would have come out independent of υ. It would only have been necessary to take

$$\Upsilon(\mathbf{x}) = \left[\int_{\mathbf{x}'}^{\mathbf{x}} \frac{d\mathbf{x}}{\mathbf{f}^{(0)}}\right] \Big/ S(\mathbf{x}'), \tag{B16}$$

where $\mathbf{x}'$ is some definite point on the loop through $\mathbf{x}$ (say the intersection with the hypersurface suggested before), S is the period (Sec. A6), and the line integral is taken along the loop (so that the ratio of vectors under the integral sign is well defined and scalar and, to be sure, nothing but ds). In carrying out our procedure in particular problems, it is in fact very desirable to start with such an Υ. We have not specified this to-be-so-called *judicious choice*, however, since for our abstract purposes it would make little difference, and that of dubious advantage; but we shall occasionally note where it would affect matters.

6. Nice Variables

The defects we have noticed in (B13) are all due to the dependence of $\mathbf{g}$ and ψ on υ. We therefore ask whether we can find an infinitely differentiable formal transformation to new variables which are similar to $\mathbf{y}$ and υ and satisfy equations similar to (B13), but without the odious dependence on the angle-like variable—and not only to lowest order but to all orders. We shall see that the answer is yes.

Denote the new sought variables by a vector $\mathbf{z}$ of $N - 1$ components and an angle-like variable ϕ, related to the old variables by

$$\mathbf{z} = \mathbf{Z}(\mathbf{y}, \upsilon), \qquad \phi = \Phi(\mathbf{y}, \upsilon), \tag{B17}$$

where $\mathbf{Z}$ and Φ are to be determined. Let the system (B13), and indirectly therefore (B4), transform into

$$\mathbf{z}_s = \epsilon\, \mathbf{h}(\mathbf{z}), \qquad \phi_s = \omega(\mathbf{z}), \tag{B18}$$

analogous to (B13) but without ϕ appearing as an argument on the right-hand sides; $\mathbf{h}$ and ω are also to be determined. Naturally $\mathbf{h}$ and ω, like $\mathbf{g}$ and ψ, will be infinite power series in ϵ, and it is only natural that $\mathbf{Z}$ and Φ be so too. Thus, the transformation (B17), unlike (B11), depends on the expansion parameter ϵ. But this is of no concern, especially since $\mathbf{Z}$ and Φ, and for that matter $\mathbf{h}$ and ω, are to be $O(1)$.

We must impose on $\mathbf{Z}$ and Φ periodicity conditions which express that υ is an angle-like variable and that ϕ is to be a similar one, namely,

$$\begin{aligned} \mathbf{Z}(\mathbf{y}, \upsilon + 1) &= \mathbf{Z}(\mathbf{y}, \upsilon), \\ \Phi(\mathbf{y}, \upsilon + 1) &= \Phi(\mathbf{y}, \upsilon) + 1. \end{aligned} \tag{B19}$$

It turns out that we may specify the dependence of $\mathbf{Z}$ and Φ on $\mathbf{y}$ for $\upsilon = 0$ with great freedom; for definiteness and simplicity we choose the initial conditions

$$\mathbf{Z}(\mathbf{y}, 0) = \mathbf{y}, \qquad \Phi(\mathbf{y}, 0) = 0. \tag{B20}$$

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These express that $\mathbf{z}$ and ϕ are to equal $\mathbf{y}$ and v at one definite point on each loop, where $v = 0$.

With conditions (B18), (B19), and (B20) we now claim not only the existence but also the uniqueness of the formal transformation we are seeking.

7. Determination of Recursion Relations

The first step in determining $\mathbf{Z}$, Φ, $\mathbf{h}$, and ω is to substitute (B17) in (B18), use the chain rule of differentiation, and eliminate $\mathbf{y}_s$ and v_s by (B13), which yields

$$\epsilon\, \mathbf{Z}_{\mathbf{y}}\cdot\mathbf{g} + \mathbf{Z}_v\psi = \epsilon\, \mathbf{h}(\mathbf{Z}), \tag{B21}$$

$$\epsilon\, \Phi_{\mathbf{y}}\cdot\mathbf{g} + \Phi_v\psi = \omega(\mathbf{Z}). \tag{B22}$$

Let us assume temporarily that we know $\mathbf{h}$ and ω as functions of their arguments. These equations may then be construed, thanks to the factor ϵ appearing in the first term of each of them, as determining, order by order, the derivatives of $\mathbf{Z}$ and Φ with respect to v in terms of known quantities, while $\mathbf{y}$ plays the role of a parameter. Accordingly, we transfer the first terms to the right-hand side, divide by ψ, and integrate; in view of the initial conditions (B20) the results may be written

$$\mathbf{Z} = \mathbf{y} + \epsilon \int_0^v dv\, [\mathbf{h}(\mathbf{Z}) - \mathbf{Z}_{\mathbf{y}}\cdot\mathbf{g}]\, \frac{1}{\psi}, \tag{B23}$$

$$\Phi = \int_0^v dv\, [\omega(\mathbf{Z}) - \epsilon\, \Phi_{\mathbf{y}}\cdot\mathbf{g}]\, \frac{1}{\psi}. \tag{B24}$$

On these we next wish to impose the periodicity conditions (B19), which it suffices to use only for $v = 0$ [for, assuming that (B19) are satisfied in the previous orders, the integrands of (B23) and (B24) are periodic functions of v]. We obtain

$$\int_0^1 dv\, [\mathbf{h}(\mathbf{Z}) - \mathbf{Z}_{\mathbf{y}}\cdot\mathbf{g}]\, \frac{1}{\psi} = 0, \tag{B25}$$

$$\int_0^1 dv\, [\omega(\mathbf{Z}) - \epsilon\, \Phi_{\mathbf{y}}\cdot\mathbf{g}]\, \frac{1}{\psi} = 1. \tag{B26}$$

These are the conditions we employ to determine the functions $\mathbf{h}$ and ω, which we previously assumed known.

To this end, denote the integral in (B23) by $\mathbf{K}$, substitute $\mathbf{y} + \epsilon\mathbf{K}$ for the argument of $\mathbf{h}$ in (B25) and of ω in (B26), and expand $\mathbf{h}$ and ω in Taylor series around the argument $\mathbf{y}$. The factors $\mathbf{h}(\mathbf{y})$ and $\omega(\mathbf{y})$ in the leading terms of these expansions, being independent of v, come out of the integrals, and we obtain

$$\mathbf{h}(\mathbf{y}) \int_0^1 dv\, \frac{1}{\psi} = \int_0^1 dv \left[\mathbf{Z}_{\mathbf{y}}\cdot\mathbf{g} - \epsilon \left\{ \frac{\mathbf{h}(\mathbf{y} + \epsilon\mathbf{K}) - \mathbf{h}(\mathbf{y})}{\epsilon} \right\} \right] \frac{1}{\psi}, \tag{B27}$$

$$\omega(\mathbf{y}) \int_0^1 dv\, \frac{1}{\psi} = 1 + \epsilon \int_0^1 dv \left[\Phi_{\mathbf{y}}\cdot\mathbf{g} - \left\{ \frac{\omega(\mathbf{y} + \epsilon\mathbf{K}) - \omega(\mathbf{y})}{\epsilon} \right\} \right] \frac{1}{\psi}; \tag{B28}$$

the expressions enclosed in curly brackets are to be thought of as expanded in the obvious way, and are clearly $O(1)$. The factor $\int_0^1 dv/\psi$ can be developed as a series with a leading term which is $O(1)$ and definitely positive, due to the nature of ψ. Thus (B27) and (B28) provide recursion formulas for $\mathbf{h}(\mathbf{y})$ and $\omega(\mathbf{y})$.

8. Recursive Construction of Desired Functions

We are now prepared to prove the existence and uniqueness of formal series $\mathbf{Z}(\mathbf{y}, v)$, $\Phi(\mathbf{y}, v)$, $\mathbf{h}(\mathbf{y})$, and $\omega(\mathbf{y})$ satisfying (B17), (B18), (B19), and (B20). (Since it is $\mathbf{h}$ and ω as functions that are sought, it is of no account that we now denote their arguments by $\mathbf{y}$ instead of $\mathbf{z}$.) We proceed by induction. It suffices to prove, for each non-negative integer n, that if the series exist and are unique up to terms of order $n - 1$, then they do and are so up to terms of order n. (For they certainly exist and are unique up to order minus one, since they are required to start with terms of order zero!)

The argument is easy; we need merely appeal to the determining Eqs. (B23), (B24), (B27), (B28) we have derived, making sure to find $\mathbf{Z}$ before Φ or $\mathbf{h}$, and ω before Φ. Thus, we may first uniquely determine $\mathbf{Z}$ to order n by (B23), noting that the right-hand side can be obtained to that order in terms of lower order quantities, already known according to the induction hypothesis. Of course, it is supposed that $\mathbf{h}(\mathbf{Z})$ has been expanded in a Taylor series around its zeroth-order argument $\mathbf{y}$, the products and quotients expanded out, and the integration performed. Similarly, we may then uniquely obtain $\mathbf{h}$ from (B27), ω from (B28), and finally Φ from (B24).

As a matter of fact, since (B23) and (B27) do not involve Φ or ω, we can find $\mathbf{Z}$ and $\mathbf{h}$ to all orders before finding Φ and ω at all.

The construction of $\mathbf{Z}$ and Φ guarantees trivially that they satisfy the initial conditions (B20) to order n. At the same time, the construction of $\mathbf{h}$ and ω guarantees that (B25) and (B26) are satisfied to order n, and therefore, likewise the periodicity conditions (B19) for $v = 0$. But $\mathbf{Z}v$ as determined by (B21) is clearly periodic in v up to order n, since $\mathbf{Z}$ is periodic up to order $n - 1$ by induction hypothesis, and so the first periodicity condition holds in general. Similarly, it then follows that Φv is also periodic up to order n and hence that the second

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periodicity condition holds in general. This completes the proof.

It is obvious by our construction that the formal series transformation from $\mathbf{y}$ and υ to $\mathbf{z}$ and ϕ expressed by (B17) is infinitely differentiable, and we can easily show that it possesses an infinitely differentiable inverse transformation. It is only necessary to obtain recursion relations for finding $\mathbf{y}$ and υ order by order, and the proof proceeds like the one just completed. The formula for $\mathbf{y}$ is merely (B23) with its last term shifted to the other side, and with the judicious choice we could have obtained the one for υ equally simply. As it is, we observe from (B24) and (B28) and from the positiveness of ψ that the function which gives ϕ to zeroth order as a function of $\mathbf{y}$ and υ can be inverted to give υ to zeroth order as a function of ϕ and $\mathbf{y}$ and hence of ϕ and $\mathbf{z}$; applying this inverse function to (B24) [i.e., using each side of (B24) in turn as argument of this inverse function and equating the results] and expanding leads to an equation whose only term of zeroth order on the right is just υ, which may then be solved for trivially to give the desired recursion relation.

9. Restatement of Formal Result

Let us restate what has been achieved, omitting the intermediate variables $\mathbf{y}$, υ and redefining $\mathbf{Z}$, Φ, and $\mathbf{X}$ accordingly. We have shown that, given the original infinitely differentiable system (B4) with all solutions nearly periodic, it is possible to find a transformation

$$\mathbf{z} = \mathbf{Z}(\mathbf{x}), \qquad \phi = \Phi(\mathbf{x}) \tag{B29}$$

and functions $\mathbf{h}(\mathbf{z})$, $\omega(\mathbf{z})$ such that the new variables satisfy (B18); moreover, the transformation possesses an inverse

$$\mathbf{x} = \mathbf{X}(\mathbf{z}, \phi), \tag{B30}$$

and $\mathbf{Z}$, Φ, $\mathbf{h}$, ω, and $\mathbf{X}$ are infinitely differentiable infinite series of non-negative powers of ϵ (the zeroth-order term of ω being definitely positive).

10. Proof That Series Represents True Solution Asymptotically

We now wish to prove that the solutions of (B18) provide asymptotically correct solutions of (B4) and also of any true system (B1) of which (B4) is the formal expansion. What this means, roughly speaking, is that if we are seeking the solution of (B1) or (B4) with a prescribed initial value at $s = 0$, and are content with an approximation to within some particular positive power of ϵ, then we may solve correspondingly approximate versions of (B18) instead, with initial values for $\mathbf{z}$ and ϕ obtained from that for $\mathbf{x}$ by a correspondingly approximate version of the transformation (B29), and convert the solutions so obtained into $\mathbf{x}$ space by a correspondingly approximate version of the inverse transformation (B30). And the claim is that we thereby obtain the desired approximate solution of (B1) or (B4), not only for s bounded but even for $s = O(1/\epsilon)$.

Owing to our desire for results valid for such large s, we must initially employ approximations up to two orders better than that of our conclusion, or than we would have needed to obtain the same conclusion restricted to bounded values of s. This is because we lose an order of accuracy, so to speak, in integrating over a large range of s, which we do twice. It is the estimation of the mobile phase angle that necessitates much of the extra accuracy; the drift of $\mathbf{z}$ alone is easier to follow, as will become clear.

To state precisely what we shall prove, let $\mathbf{x}^\dagger(s)$ be a solution of (B1) or alternatively of the system obtained from (B4) by truncating the formal series at order $n + 1$ ($n \geq 0$), so that

$$\mathbf{x}^\dagger_s = \mathbf{F}(\mathbf{x}^\dagger, \epsilon) \quad \text{or} \quad \mathbf{x}^\dagger_s = \mathbf{f}^{[n+1]}(\mathbf{x}^\dagger); \tag{B31}$$

these are of course genuine systems of differential equations, and not merely formal. Define $\mathbf{z}'(s)$ and $\phi'(s)$ as those solutions of the equations obtained from (B18) by appropriate analogous truncations of $\mathbf{h}$ and ω which have at $s = 0$ the values obtained from $\mathbf{x}^\dagger(0)$ by an appropriately truncated version of transformation (B29); specifically, $\mathbf{z}'(s)$ and $\phi'(s)$ are uniquely defined by the conditions

$$\mathbf{z}'_s = \epsilon\, \mathbf{h}^{[n]}(\mathbf{z}'), \qquad \phi'_s = \omega^{[n]}(\mathbf{z}'), \tag{B32}$$

$$\mathbf{z}'(0) = \mathbf{Z}^{[n]}(\mathbf{x}^\dagger(0)), \qquad \phi'(0) = \Phi^{[n]}(\mathbf{x}^\dagger(0)). \tag{B33}$$

Next, transform to obtain a curve in $\mathbf{x}$ space, defining $\mathbf{x}'(s)$ as the result of applying the appropriately truncated version of transformation (B30) to $\mathbf{z}'(s)$ and $\phi'(s)$:

$$\mathbf{x}' = \mathbf{X}^{[n-1]}(\mathbf{z}', \phi'). \tag{B34}$$

The claim now is that $\mathbf{x}'$ is a good approximation to $\mathbf{x}^\dagger$, specifically that

$$\mathbf{x}^\dagger(s) = \mathbf{x}'(s) + O(\epsilon^n), \tag{B35}$$

for s within a range of order $1/\epsilon$, so long as $\mathbf{x}^\dagger$ and $\mathbf{x}'$ stay in the domain of definition of $\mathbf{f}$. [For all we know yet this might limit s more severely, but in the next section it will be shown that they do stay in for $s = O(1/\epsilon)$.]

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The proof proceeds in three stages. First we show that $\mathbf{Z}^{[n+1]}(\mathbf{x}^\dagger)$ is close to $\mathbf{z}'$, then that $\Phi^{[n]}(\mathbf{x}^\dagger)$ is close to ϕ', and finally that $\mathbf{x}'$ is close to $\mathbf{x}^\dagger$, as desired. We need the three formal identities in $\mathbf{x}$

$$\mathbf{Z}_\mathbf{x}\cdot\mathbf{f} = \epsilon\,\mathbf{h}(\mathbf{Z}), \tag{B36}$$

$$\Phi_\mathbf{x}\cdot\mathbf{f} = \omega(\mathbf{Z}), \tag{B37}$$

$$\mathbf{x} = \mathbf{X}(\mathbf{Z}, \Phi), \tag{B38}$$

the first two of which express that (B29) transforms (B4) into (B18), and the last that (B30) inverts (B29). Or rather, we need their approximate versions

$$\mathbf{Z}_\mathbf{x}^{[n+1]}\cdot\mathbf{f}^{[n+1]} = \epsilon\,\mathbf{h}^{[n]}(\mathbf{Z}^{[n+1]}) + O(\epsilon^{n+2}), \tag{B39}$$

$$\Phi_\mathbf{x}^{[n]}\cdot\mathbf{f}^{[n+1]} = \omega^{[n]}(\mathbf{Z}^{[n]}) + O(\epsilon^{n+1}), \tag{B40}$$

$$\mathbf{x} = \mathbf{X}^{[n-1]}(\mathbf{Z}^{[n-1]}, \Phi^{[n-1]}) + O(\epsilon^n), \tag{B41}$$

the last of which, for instance, may be shown as follows: The difference between the left-hand side and the first term on the right is a well-defined infinitely differentiable function of $\mathbf{x}$ and ϵ. Its Taylor expansion around $\epsilon = 0$ vanishes identically up to terms of order $n - 1$, since it contains exactly the same terms up to that order as occur in the formal identity (B38). Thus, the quotient of the difference by ϵ^n is a continuous function of $\mathbf{x}$ and ϵ on a closed bounded domain and is therefore bounded independently of $\mathbf{x}$ and ϵ, which completes the proof. The same argument, but with n replaced everywhere by $n + 2$ or $n + 1$, applies to (B39) and (B40).

If we temporarily restrict ourselves to the second alternative in (B31), then (B39) evaluated at $\mathbf{x}^\dagger$ may be written

$$\{\mathbf{Z}^{[n+1]}(\mathbf{x}^\dagger)\}_s = \epsilon\,\mathbf{h}^{[n]}(\mathbf{Z}^{[n+1]}(\mathbf{x}^\dagger)) + O(\epsilon^{n+2}). \tag{B42}$$

We now wish to compare this with the first equation (B32), and invoke the approximation theorem of Appendix 1. The crux of the proof is to change the independent variable first, setting

$$\sigma = \epsilon\, s. \tag{B43}$$

The two equations we wish to compare become thereby (after division by ϵ)

$$\{\mathbf{Z}^{[n+1]}(\mathbf{x}^\dagger)\}_\sigma = \mathbf{h}^{[n]}(\mathbf{Z}^{[n+1]}(\mathbf{x}^\dagger)) + O(\epsilon^{n+1}), \tag{B44}$$

$$\mathbf{z}'_\sigma = h^{[n]}(\mathbf{z}'). \tag{B45}$$

Thus $\mathbf{Z}^{[n+1]}(\mathbf{x}^\dagger)$ and $\mathbf{z}'$ satisfy the same autonomous system of differential equations within $O(\epsilon^{n+1})$, and since by (B33) they also have the same initial values within $O(\epsilon^{n+1})$, according to the approximation theorem

$$\mathbf{Z}^{[n+1]}(\mathbf{x}^\dagger) = \mathbf{z}' + O(\epsilon^{n+1}) \tag{B46}$$

for bounded σ, i.e., for $s = O(1/\epsilon)$. (In this, we might as well replace the truncation index $n + 1$ on the left by n.) This completes the first stage, except for the remark that the case of the first alternative in (B31) may be treated in exactly the same way by using, instead of (B39), the obviously equally valid formula obtained from it by replacing the factor $\mathbf{f}^{[n+1]}$ by $\mathbf{F}$.

The second stage is even simpler. Restricting ourselves again to the second alternative (with the same trivial modification to cover the first alternative), we evaluate (B40) at $\mathbf{x}^\dagger$ and write it in the form

$$\{\Phi^{[n]}(\mathbf{x}^\dagger)\}_s = \omega^{[n]}(\mathbf{Z}^{[n]}(\mathbf{x}^\dagger)) + O(\epsilon^{n+1})$$
$$= \omega^{[n]}(\mathbf{z}') + O(\epsilon^{n+1}) = \phi'_s + O(\epsilon^{n+1}), \tag{B47}$$

where we have used (B46) and the second Eq. (B32). In view of the common initial values, asserted in the second Eq. (B33), a simple integration immediately yields

$$\Phi^{[n]}(\mathbf{x}^\dagger) = \phi' + O(\epsilon^n) \tag{B48}$$

for $s = O(1/\epsilon)$. (Again, we might as well replace the truncation index n on the left by $n - 1$.)

The third stage is the simplest of all. Evaluating (B41) at $\mathbf{x}^\dagger$ and using (B46) and (B48) yields

$$\mathbf{x}^\dagger = \mathbf{X}^{[n-1]}(\mathbf{z}', \phi') + O(\epsilon^n), \tag{B49}$$

which, in view of (B34), is precisely (B35). The proof is complete.

11. Proof that the True Solution Remains in Domain for Many Gyrations

We can now demonstrate that $\mathbf{x}^\dagger(s)$ actually remains interior to the given domain of $\mathbf{x}$ space over a large range in s of order $1/\epsilon$. For (B45) shows that there is some finite range of σ over which $\mathbf{z}'$ stays interior to the corresponding domain of $\mathbf{z}$ space. This corresponds by (B43) to some definite large range of s, to which we restrict the discussion from now on. [Since ϕ' is an angle-like variable it is subject to no restriction in magnitude, and to be sure it varies by $O(1/\epsilon)$, as we see from (B32).] From (B34) it is then clear that $\mathbf{x}'$ is defined and interior to the domain, so (B35) holds so long as $\mathbf{x}^\dagger$ stays inside. On the other hand, so long as (B35) holds, $\mathbf{x}^\dagger$ does in fact stay in. Pulling ourselves up by the bootstraps in this way, we arrive at the conclusion that $\mathbf{x}^\dagger$ stays inside as long as $\mathbf{x}'$ does (at least very nearly, though naturally one may reach the boundary a little before the other).

If this argument does not appear immediately

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convincing (as may well be the case), it can be elaborated as follows. Knowing that $\mathbf{x}'$ stays well within the domain for some range of s of order $1/\epsilon$, that is, stays a finite distance δ away from the boundary, let us assume (for the sake of a *reductio ad absurdum*) that $\mathbf{x}^\dagger$ leaves the domain during that range. Consider the value of s nearest to zero for which $\mathbf{x}^\dagger$ and $\mathbf{x}'$ are at a distance of $\delta/2$. Since $\mathbf{x}^\dagger$ and $\mathbf{x}'$ are both within the domain up to that point, (B35) applies and shows that they are separated by only $O(\epsilon^n)$, which is much less than $\delta/2$. This contradiction proves that $\mathbf{x}^\dagger$, in fact, could not have left the domain during the range of interest.

C. RINGS AND THE REDUCED SYSTEM

1. Theorem of Phase Independence

At this point we introduce a simple but powerful tool, which we shall apply repeatedly to establish that functions are independent of the phase angle ϕ. Let $\mathbf{W}(\mathbf{z}, \phi)$ be a vector with any finite number M of components, a formal infinite series in powers of ϵ, $O(1)$, and a function of its indicated arguments $\mathbf{z}$ and ϕ, periodic in the latter (of course with period unity). Let $\mathbf{W}$ satisfy a formal differential equation

$$\mathbf{W}_\phi = \mathbf{A}(\mathbf{z}) + \epsilon\, \mathbf{B}(\mathbf{z}, \mathbf{W}, \mathbf{W}_\mathbf{z}), \tag{C1}$$

where $\mathbf{A}(\mathbf{z})$ and $\mathbf{B}(\mathbf{z}, \mathbf{w}, \mathbf{v})$ are vectors of M components, formal infinite series in powers of ϵ, and functions of their indicated arguments [$\mathbf{w}$ being a generic variable with M components, and $\mathbf{v}$ one with $M(N-1)$ components], and where $\mathbf{B}$ is $O(1)$, but no such assumption is made about $\mathbf{A}$. Then $\mathbf{W}$ is in fact independent of ϕ to all orders.

To prove this *theorem of phase independence*, we merely form the indefinite integral of (C1),

$$\mathbf{W}(\mathbf{z}, \phi) = \mathbf{W}(\mathbf{z}, 0) + \int_0^\phi d\phi\, [\mathbf{A}(\mathbf{z}) + \epsilon\, \mathbf{B}(\mathbf{z}, \mathbf{W}, \mathbf{W}_\mathbf{z})]; \tag{C2}$$

impose the periodicity condition on $\mathbf{W}$,

$$\int_0^1 d\phi\, [\mathbf{A}(\mathbf{z}) + \epsilon\, \mathbf{B}(\mathbf{z}, \mathbf{W}, \mathbf{W}_\mathbf{z})] = 0; \tag{C3}$$

and proceed by induction. Up to order minus one $\mathbf{W}$ vanishes and so is independent of ϕ. Suppose $\mathbf{W}$ is independent of ϕ up to terms of order $n-1$. Up to order n, then, the integrand of (C3) is independent of ϕ, and so equals its own integral, which vanishes. Therefore (C2) may be written

$$\mathbf{W}(\mathbf{z}, \phi) = \mathbf{W}(\mathbf{z}, 0) + O(\epsilon^{n+1}), \tag{C4}$$

which shows that $\mathbf{W}$ is independent of ϕ up to terms of order n, and thereby completes the induction. (And we have as a side result that $\mathbf{A} = O(\epsilon)$, and in fact that $\mathbf{A} + \epsilon\mathbf{B} = 0$ to all orders.)

It is clear that we could have allowed $\mathbf{B}$ to depend on higher derivatives of $\mathbf{W}$ with respect to $\mathbf{z}$, with no change in the argument or the conclusion. Of this we shall make no use, but there is a slight modification of the theorem which we shall use very frequently. Namely, let us specialize by assuming that $\mathbf{B}$ is a linear function of its last two arguments. We can then trivially generalize by allowing $\mathbf{W}$ to start with any (e.g., a negative) power of ϵ, instead of being $O(1)$, and obviously the same conclusion holds.

Another slight but useful modification is obtained by replacing $\mathbf{W}_\phi$ by $\mathbf{W}_s$ in (C1). Since

$$\mathbf{W}_s = \mathbf{W}_\phi\, \omega(\mathbf{z}) + \epsilon\, \mathbf{W}_\mathbf{z} \cdot \mathbf{h}(\mathbf{z}), \tag{C5}$$

the resulting equation can be immediately put back in the same form as (C1) with $\mathbf{A}$ and $\mathbf{B}$ appropriately modified, so the conclusion still holds.

It is convenient to observe here, what will be needed later and is obvious from (C5), that the operators of differentiation with respect to s and to ϕ commute with each other.

2. Arbitrariness of Nice Variables

It is clear that the nice variables $\mathbf{z}$ and ϕ obtained in part B are far from unique. For, the process by which we determined them involved some arbitrary choices, namely of the functions $\mathbf{Y}$ and Υ in Sec. B.5 and of the form of the initial conditions (B20). We address ourselves now to the investigation of exactly how much freedom there is finally in the choice of such desirable coordinates.

Suppose that we were to introduce new variables $\mathbf{z}^\dagger$ and $\phi^\dagger$, expressed as infinitely differentiable formal infinite series in non-negative powers of ϵ, with coefficients depending on $\mathbf{z}$ and ϕ, such that $\mathbf{z}^\dagger$ is periodic in ϕ and $\phi^\dagger$ changes by unity with ϕ, and such that the transformation is formally invertible to give similar expressions for $\mathbf{z}$ and ϕ in terms of $\mathbf{z}^\dagger$ and $\phi^\dagger$. We then may ask, first, under what reasonably general conditions $\mathbf{z}^\dagger$ and $\phi^\dagger$ will necessarily satisfy equations analogous to (B18).

An answer is easy to find. For $\mathbf{z}$ satisfies an autonomous system of equations, and this will certainly transform into an autonomous system for $\mathbf{z}^\dagger$ so long as $\mathbf{z}^\dagger$ is obtained by a transformation from $\mathbf{z}$ alone, i.e., so long as

$$\mathbf{z}^\dagger_\phi = 0. \tag{C6}$$

[Indeed, it is then immediate that $\mathbf{z}^\dagger_s = \epsilon\, \mathbf{h}^\dagger(\mathbf{z}^\dagger)$,

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where $\mathbf{h}^\dagger(\mathbf{z}^\dagger) = \mathbf{z}^\dagger_{\mathbf{z}} \cdot \mathbf{h}(\mathbf{z})$.] Furthermore, ϕ may be thought of as satisfying the trivial first-order differential equation $\phi_s = \omega(\mathbf{z})$ if $\mathbf{z}$ is considered a parameter, and $\phi^\dagger$ will certainly satisfy a similar trivial equation so long as it differs from ϕ by only a "constant," which may of course depend on the "parameter" $\mathbf{z}$ (we cannot take $\phi^\dagger$ to be a more general linear function of ϕ, since the two angle variables must change by unity together), namely, so long as

$$\phi^\dagger_\phi = 1. \tag{C7}$$

[Indeed, it is then immediate that $\phi^\dagger_s = \omega^\dagger(\mathbf{z}^\dagger)$, where $\omega^\dagger(\mathbf{z}^\dagger) = \epsilon\, \phi^\dagger_{\mathbf{z}} \cdot \mathbf{h}(\mathbf{z}) + \omega(\mathbf{z})$; note that $\phi^\dagger_{\mathbf{z}}$ depends on $\mathbf{z}$ alone, since $\phi^\dagger_{\phi\mathbf{z}} = 0$.]

We next may ask, conversely, whether every transformation from $\mathbf{z}$, ϕ to new nice variables $\mathbf{z}^\dagger$, $\phi^\dagger$ in fact satisfies (C6) and (C7). The answer is yes; the conditions we have found are necessary as well as sufficient to preserve the form of (B18). For to begin with, suppose there exists a function $\mathbf{h}^\dagger$ such that

$$\mathbf{z}^\dagger_s = \epsilon\, \mathbf{h}^\dagger(\mathbf{z}^\dagger); \tag{C8}$$

then we may apply the theorem of phase independence and deduce (C6). And suppose also that there exists a function $\omega^\dagger$ such that

$$\phi^\dagger_s = \omega^\dagger(\mathbf{z}^\dagger); \tag{C9}$$

to this we cannot directly apply the theorem of phase independence (even though we know that $\mathbf{z}^\dagger$ depends on $\mathbf{z}$ only), because $\phi^\dagger$ is not periodic in ϕ, but we may rewrite (C9) as

$$[\phi^\dagger - \phi]_s = \omega^\dagger(\mathbf{z}^\dagger) - \omega(\mathbf{z}), \tag{C10}$$

to which we can, because $\phi^\dagger - \phi$ is periodic; we thereby deduce that

$$[\phi^\dagger - \phi]_\phi = 0, \tag{C11}$$

which is equivalent to (C7).

3. Rings, Phase Differences, and the Reduced System

The necessary and sufficient conditions (C6) and (C7) that new variables $\mathbf{z}^\dagger$, $\phi^\dagger$ be as nice as $\mathbf{z}$, ϕ enable us to introduce some simple concepts connected with the original system (B4) in $\mathbf{x}$ which are based on the existence of nice variables, but are independent of which ones they specifically are. These concepts will of course be defined only to all orders, and not in general exactly.

Consider the set of all points in $\mathbf{x}$ space which correspond to a common value of $\mathbf{z}$. Since ϕ is an angle variable, this set is a topological circle which we call a *ring*. Two points in $\mathbf{x}$ space with the same $\mathbf{z}$ we call *ringmates*. Now (C6) has the simple interpretation that $\mathbf{z}^\dagger$ is constant around any ring, which shows that the concept of rings and ringmates is invariant with respect to the choice of nice variables.

The particular value of the phase variable ϕ corresponding to a point $\mathbf{x}$ has of course no invariant significance. But if two points are ringmates, then their *phase difference* $\Delta\phi$ does have, for, by (C7),

$$\Delta\phi^\dagger = \int d\phi^\dagger = \int \phi^\dagger_\phi\, d\phi = \int d\phi = \Delta\phi, \tag{C12}$$

where the integrals are extended over a portion of the ring between the two points. Of course this phase difference is only defined modulo unity.

It is also clear that the knowledge of which points are ringmates and of the phase differences between those that are is all the information inherent in knowing some nice variables but independent of which ones they are. For from that knowledge one can construct nice variables.

Since $\mathbf{z}$ satisfies an autonomous system, it is obvious that if two solutions $\mathbf{x}(s)$ of the original system are ringmates at any one value of s, then they are so at every value of s. Also, their phase difference is always the same, since $(\Delta\phi)_s = \omega(\mathbf{z}) - \omega(\mathbf{z}) = 0$. It is therefore meaningful to apply the invariant concepts of rings and of phase difference not only to points in $\mathbf{x}$ space but also to entire solutions $\mathbf{x}(s)$ of (B4).

The ring constitutes a sophisticated or refined version of the loop, which is indeed a zeroth-order approximation to it. Each ring is characterized by its value of $\mathbf{z}$, just as each loop was characterized (in Sec. B.5) by its value of $\mathbf{y}$. That is, the rings form an $(N-1)$-dimensional space, and their slow drift as s varies is governed by the formal autonomous system

$$\mathbf{z}_\sigma = \mathbf{h}(\mathbf{z}), \tag{C13}$$

where we have reintroduced the change of independent variable (B43) to $\sigma = \epsilon s$ in order that what we shall call the *reduced system* (C13) might be in the same general form as the original system (B4) (though not necessarily in standard form). The phase now plays a subordinate role (in contrast to its previous rather dominant one), since it has no effect (to any finite order) on the drift.

Among other things, we have shown that the nearly recurrent system (B4) is splittable. The new

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autonomous system $\mathbf{z}_s = \epsilon\,\mathbf{h}(\mathbf{z})$ contains one less variable.

If (C13) should itself happen to be nearly recurrent and in standard form, we can reapply the whole procedure to it and obtain a doubly reduced system, and so on. The arguments of Sec. B.11 can then be extended to show that the true solution $\mathbf{x}^\dagger(s)$ of (B31) remains in the relevant domain of $\mathbf{x}$, space for a range of s even larger than $O(1/\epsilon)$.

4. Roto-rate

It should at this point be apparent that all the information contained in knowledge of the rings and of phase differences on them is contained, in "capsule" form, in knowledge of $\mathbf{x}_\phi$. For, as we show in detail shortly, the rings are obtainable from $\mathbf{x}_\phi$ by integration of an autonomous system whose independent variable is the phase. It is noteworthy that in terms of $\mathbf{x}_\phi$ as a function[18] of $\mathbf{x}$ we may express our results entirely within $\mathbf{x}$ space, even though it was convenient to transform to other variables in deriving them; so expressed, furthermore, the results take on an esthetically satisfying uniqueness, since for any other nice coordinates $\mathbf{z}^\dagger$, $\phi^\dagger$ we have

$$\mathbf{x}_\phi = \mathbf{x}_{\mathbf{z}\dagger}\cdot\mathbf{z}^\dagger_\phi + \mathbf{x}_{\phi\dagger}\,\phi^\dagger_\phi = \mathbf{x}_{\phi\dagger} \tag{C14}$$

by (C6) and (C7). The function merits a specific name and symbol, and so we define the *roto-rate* $\mathbf{R}(\mathbf{x})$ by

$$\mathbf{R} \equiv \mathbf{x}_\phi, \tag{C15}$$

the name being supposed to suggest the defining characteristic of $\mathbf{R}$ as the rate of change (with respect to phase) of the point at $\mathbf{x}$ in rotating around its ring.

We next derive four properties of $\mathbf{R}$ which can be expressed independently of nice coordinates, rings, and phases, and show afterward that these properties characterize it uniquely. The first property is that the autonomous system (C15) is recurrent; this is obvious because every solution in fact runs around a ring. The second property (also obvious) is that the initial value recurs (for the first time) when the independent variable ϕ has increased by unity; this may be expressed by writing

$$\oint \frac{d\mathbf{x}}{\mathbf{R}} = 1, \tag{C16}$$

where the line integral is taken once around any integral curve of $\mathbf{R}$, and the ratio of vectors is defined, since precisely on such a curve the vectors are parallel (and, in fact, $d\mathbf{x} = \mathbf{x}_\phi\,d\phi = \mathbf{R}\,d\phi$). The third property is that to lowest (zeroth) order, $\mathbf{R}$ is parallel to $\mathbf{f}$ and unequal to zero, as seen immediately from (B4) and from (C5) with $\mathbf{W}$ replaced by $\mathbf{x}$. Finally, the fourth property is that

$$\mathbf{f}_\mathbf{x}\cdot\mathbf{R} = \mathbf{R}_\mathbf{x}\cdot\mathbf{f}, \tag{C17}$$

which expresses the consistency of (B4) with (C15) and is, in fact, a trivial rewriting of

$$\mathbf{f}_\mathbf{x}\cdot\mathbf{x}_\phi = \mathbf{f}_\phi = \mathbf{x}_{s\phi} = \mathbf{x}_{\phi s} = \mathbf{R}_s = \mathbf{R}_\mathbf{x}\cdot\mathbf{x}_s, \tag{C18}$$

in which we have used the observation at the end of Sec. C.1.

To show that the roto-rate is uniquely characterized by these four properties, it suffices to show that for any (infinitely differentiable formal series) function $\mathbf{R}$ with these properties we can find nice variables $\mathbf{z}$ and ϕ such that (C15) holds. This is because for any other function $\mathbf{R}^\dagger$ with the four properties we could then find nice variables $\mathbf{z}^\dagger$ and $\phi^\dagger$ such that $\mathbf{R}^\dagger = \mathbf{x}_{\phi\dagger}$, whereupon (C14) would show immediately that $\mathbf{R} = \mathbf{R}^\dagger$.

We may construct the desired nice variables as solutions of

$$\mathbf{z}_\mathbf{x}\cdot\mathbf{R} = 0, \tag{C19}$$

$$\phi_\mathbf{x}\cdot\mathbf{R} = 1; \tag{C20}$$

that is, the $\mathbf{z}$ are constructed to be (of course independent) constants on the closed (first property!) integral curves of $\mathbf{R}$, and ϕ to be multi-valued. That ϕ is an angle variable, i.e., that its branches differ by integers, follows from (C16) (second property!),

$$\oint d\phi = \oint \phi_\mathbf{x}\cdot d\mathbf{x} = \oint \phi_\mathbf{x}\cdot\mathbf{R}\,d\mathbf{x}/\mathbf{R} = 1. \tag{C21}$$

The coordinates cannot be degenerate, since the $\mathbf{z}$ were chosen independent and, if ϕ were expressible as a function of $\mathbf{z}$, we would have $\phi_\mathbf{x}\cdot\mathbf{R} = 0$ as a consequence of (C19), contradicting (C20). We may therefore employ a standard representation of the unit dyadic in the derivation

$$\mathbf{R} = \mathcal{I}\cdot\mathbf{R} = (\mathbf{x}_\mathbf{z}\cdot\mathbf{z}_\mathbf{x} + \mathbf{x}_\phi\,\phi_\mathbf{x})\cdot\mathbf{R} = \mathbf{x}_\phi, \tag{C22}$$

which verifies (C15). It is now fairly easy to establish one of the main properties of nice coordinates, that $\mathbf{z}_s$ and ϕ_s are functions of $\mathbf{z}$ only[19]:

[18] Of course, $\mathbf{x}$ is to be treated as a function of ϕ and $\mathbf{z}$ through the transformation (B30) when the differentiation is performed, but the derivative may then be treated as a function of $\mathbf{x}$ by the inverse transformation (B29).

[19] In this paragraph we cannot appeal to the commutativity of the differentiations, which would enable us to bypass several intermediate steps in (C23), because we do not yet know that $\mathbf{z}$, ϕ are nice coordinates, that being, indeed, what we wish to establish.

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$$(\mathbf{z}_s)_\phi = (\mathbf{z}_\mathbf{x}\cdot\mathbf{f})_\mathbf{x}\cdot\mathbf{R} = \mathbf{z}_{\mathbf{xx}}:\mathbf{f}\,\mathbf{R} + \mathbf{z}_\mathbf{x}\cdot\mathbf{f}_\mathbf{x}\cdot\mathbf{R}$$
$$= \mathbf{z}_{\mathbf{xx}}:\mathbf{R}\,\mathbf{f} + \mathbf{z}_\mathbf{x}\cdot\mathbf{R}_\mathbf{x}\cdot\mathbf{f} \qquad \text{(C23)}$$
$$= (\mathbf{z}_\mathbf{x}\cdot\mathbf{R})_\mathbf{x}\cdot\mathbf{f} = 0,$$

using successively (B4), (C15), the symmetry of the triadic $\mathbf{z}_{\mathbf{xx}}$ in its last two vector positions, (C17) (fourth property!), and (C19); and the same for ϕ_s. This already shows that ϕ_s may be written as $\omega(\mathbf{z})$; to show that $\mathbf{z}_s$ may be written as $\epsilon\mathbf{h}(\mathbf{z})$ it remains only to note that $\mathbf{z}_\mathbf{x}\cdot\mathbf{f} = O(\epsilon)$, in view of (C19) (third property!).

It does not appear obvious whether an explicit recursion formula to determine $\mathbf{R}$ in terms of $\mathbf{f}$ can be found. If so, the whole theory of this paper might be simplified and rendered less deep.

D. HEREDITARY PROPERTIES

1. Definition

We now have a systematic formal procedure which we can apply to any nearly recurrent formal system in N dependent variables, enabling us to "take out the fast gyration" and find a reduced system in $N - 1$ dependent variables which describes the drift of rings to all orders. There are, as we shall see, a number of properties of systems of differential equations which, if possessed by the original system (B4), are carried over or *inherited* by the reduced system (C13). We shall call a property of systems of differential equations *hereditary* if, whenever it holds for the original system (at least in the formal sense of expansions to all orders), it also holds for the reduced system. This definition, like that of local dependence, is metamathematical and need not be used with great strictness.

2. Inheritance of Splittability

As a first, fairly trivial, example of an hereditary property consider that of being splittable (Sec. A.7). If (B4) is splittable, and if the functionals which depend locally on $\mathbf{f}(\mathbf{x})$ and define the M new variables satisfying the new autonomous system are expansible in power series in ϵ, with zero-order leading terms, then the new autonomous system obviously also has all solutions nearly periodic and can be put into standard form by an appropriate change of variables. Introducing new nice variables, it can be seen by the methods of part C that the new angle variable is essentially the same as (and could have been chosen to be in fact the same as) the original angle variable ϕ, and thus that the reduced system of the new autonomous system is the new autonomous system of the original reduced system (C13). Therefore the original reduced system is splittable, and the decrease in number of dependent variables as a result of the splitting is the same as for the given splittable system (B4).

3. Inheritance of an Integral (Constant of Motion)

An important and simple hereditary property is the possession of an integral (independent of s). To see this let $I(\mathbf{x})$ be an (expansible) integral of (B4); that is, let

$$I_s = I_\mathbf{x}\cdot\mathbf{f} = 0, \qquad \text{(D1)}$$

so that I is constant on every solution of (B4). By the theorem of phase independence we immediately deduce that I is a function of $\mathbf{z}$ only, and is therefore also an integral of the reduced system (C13).

It follows from this that (B4) cannot possess a complete set of integrals (N functionally independent integrals) all of which are expansible. For (C13) can of course have only $N - 1$ independent integrals. Any integral of (B4) which distinguishes between ringmates is inexpansible.

4. Inheritance of Invariant Measure

Another interesting hereditary property is possession of an invariant measure. Let $\rho(\mathbf{x})$ be the measure density of a measure which is invariant under the "flow" represented by (B4); that is, the measure of any region in $\mathbf{x}$ space is the integral of ρ over that region, and the measures of the images of that region for different values of s under the "motion" (B4) are all equal. The mathematical expression of this invariance is just the "time-independent" equation of continuity

$$\mathcal{J} : (\rho\,\mathbf{f})_\mathbf{x} = 0, \qquad \text{(D2)}$$

where the left-hand side is nothing but the divergence of $\rho\,\mathbf{f}$, written in accordance with our subscript-differentiation notation. This may also be written

$$\rho_s + \rho\,\mathcal{J} : \mathbf{f}_\mathbf{x} = 0. \qquad \text{(D3)}$$

Such an invariant measure is intrinsically independent of coordinate system. In $\mathbf{z}$, ϕ space it has the measure density $\tau(\mathbf{z}, \phi)$ obtained from ρ by multiplying by the Jacobian of $\mathbf{x}$ with respect to $\mathbf{z}$, ϕ. We then have (D2) and (D3) in the forms

$$\mathcal{J} : (\tau\,\epsilon\,\mathbf{h})_\mathbf{z} + (\tau\,\omega)_\phi = 0, \qquad \text{(D4)}$$

$$\tau_s + \epsilon\,\tau\,\mathcal{J} : \mathbf{h}_\mathbf{z} = 0. \qquad \text{(D5)}$$

If ρ is expansible in a power series then of course so is τ (since the Jacobian certainly is), hence by the

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theorem of phase independence τ is a function of $\mathbf{z}$ alone. Thus $(\tau\omega)_\phi = 0$, and (D4) becomes for the reduced system the precise analog of (D2) for the original system, which proves the inheritance.

5. General Integral Invariants

The last two hereditary properties discussed are essentially special cases of a more general one, possession of an integral invariant[20] of any degree m. They correspond in fact to the extreme cases $m = 0$ and $m = N$, whose particular simplicity seemed to merit separate treatment. We confine ourselves in this section to the case $m = 2$, which is perhaps uniquely characterized as being enough like the general case to permit immediate generalization[21] of all concepts, results, and non-notational features, yet simple enough to permit handling in our index-free notation with no further special conventions and no particular clumsiness. However, we consider not only absolute integral invariants but also relative integral invariants.

Let $\mathfrak{A}(\mathbf{x})$ be an antisymmetric dyadic[22] field in $\mathbf{x}$ space. Then $\mathfrak{A} : d\mathbf{x}\, d\mathbf{x}$ is a second-order differential form[23] whose integral over any surface imbedded in $\mathbf{x}$ space constitutes a sort of measure for such surfaces. The rate of change of this measure with respect to s, as (each point of) the surface varies with s according to (B4), is another measure similarly associated with the differential form

$$(\mathfrak{A} : d\mathbf{x}\, d\mathbf{x})_s = \mathfrak{A}_s : d\mathbf{x}\, d\mathbf{x} + \mathfrak{A} : (d\mathbf{f}\, d\mathbf{x} + d\mathbf{x}\, d\mathbf{f}) = (\mathfrak{A}_s + \mathfrak{A}\cdot\mathbf{f}_\mathbf{x} + {}_\mathbf{x}\mathbf{f}\cdot\mathfrak{A}) : d\mathbf{x}\, d\mathbf{x}. \qquad \text{(D6)}$$

The condition that $\mathfrak{A} : d\mathbf{x}\, d\mathbf{x}$ be a *relative integral invariant* is that (D6) vanish upon integration over every closed surface, the condition for which in turn is that there exist a single-valued vector field $\mathbf{V}(\mathbf{x})$ satisfying

$$\mathfrak{A}_s + \mathfrak{A}\cdot\mathbf{f}_\mathbf{x} + {}_\mathbf{x}\mathbf{f}\cdot\mathfrak{A} = \mathbf{V}_\mathbf{x} - {}_\mathbf{x}\mathbf{V}, \qquad \text{(D7)}$$

where the right-hand side is just the curl of $\mathbf{V}$. In this case the measure of any closed surface stays constant as the surface flows. The condition that $\mathfrak{A} : d\mathbf{x}\, d\mathbf{x}$ be an *absolute integral invariant* is that (D6) vanish upon integration over every (not necessarily closed) surface, namely that the left-hand side of (D7) vanish. In this case the measure of any surface at all stays constant as the surface flows.

6. Inheritance of Relative Invariants

Transforming to nice coordinates we have

$$\mathfrak{A} : d\mathbf{x}\, d\mathbf{x} = \mathfrak{A} : (\mathbf{x}_z\cdot dz + \mathbf{x}_\phi\, d\phi)(\mathbf{x}_z\cdot dz + \mathbf{x}_\phi\, d\phi) = \mathfrak{B} : dz\, dz + \mathbf{C}\cdot(dz\, d\phi - d\phi\, dz), \qquad \text{(D8)}$$

where there can of course be no $d\phi\, d\phi$ term (and its coefficient would vanish anyway), and where

$$\mathfrak{B} \equiv {}_z\mathbf{x}\cdot\mathfrak{A}\cdot\mathbf{x}_z\,, \quad \mathbf{C} \equiv \mathbf{x}_\phi\cdot\mathfrak{A}\cdot\mathbf{x}_z\,. \qquad \text{(D9)}$$

The condition that (D8) be a relative invariant, obtainable in the same way as (D7) or by transforming (D7), is that there exist single-valued $\mathbf{W}(\mathbf{x})$ and $T(\mathbf{x})$ such that

$$\mathfrak{B}_s + \mathfrak{B}\cdot\epsilon\,\mathbf{h}_z + \epsilon\,{}_z\mathbf{h}\cdot\mathfrak{B} - \mathbf{C}\,\omega_z + \omega_z\,\mathbf{C} = \mathbf{W}_z - {}_z\mathbf{W}, \qquad \text{(D10)}$$

$$\mathbf{C}_s + \mathbf{C}\cdot\epsilon\,\mathbf{h}_z = T_z - \mathbf{W}_\phi. \qquad \text{(D11)}$$

Let us denote the complete integral over phase (equivalently, the closed line integral around a ring, or the average value with respect to ϕ) by an asterisk. Then from (D11) we have immediately (since differentiations and integrations with respect to ϕ and to s commute)

$$\mathbf{C}^*_s + \mathbf{C}^*\cdot\epsilon\,\mathbf{h}_z = T^*_z. \qquad \text{(D12)}$$

This is just the condition [analogous to (D7), but of order one less] that $\mathbf{C}^*\cdot dz$ be a relative integral invariant of the reduced system. If $\mathfrak{A}$ and $\mathbf{V}$ are expansible in ϵ then so are $\mathbf{C}$ and T and so also $\mathbf{C}^*$ and T^*. Thus the inheritance of a relative invariant is proved.

Of course $\mathbf{C}^*\cdot dz$ might happen to be a trivial relative invariant, in the sense of its integral having a constant value independent of the closed path

[20] See De Donder [Th. De Donder, *Théorie des invariants intégraux* (Gauthier-Villars, Paris, 1927)] for the definitions, for a clear exposition of the general theory of integral invariants, and accordingly for a justification of some of the steps in this section. Strictly speaking, the hereditary property is possession of an integral invariant whose degree is less than the order of the system by a definite integer, since an invariant of degree m of the original system induces one of degree $m - 1$ of the reduced system. The word "essentially" above is to indicate that from this point of view the property of Sec. 3 is slightly anomolous.

[21] The important case $m = 1$ may be even more immediately obtained by the opposite process (here "degradation" rather than specialization), and moreover is illustrated by the theory of Hamiltonian systems in part E (as also is the case $m = 2$ for a relative invariant), where the invariant has the special form $\mathbf{p}\cdot d\mathbf{q}$.

[22] In the general case it should be a completely antisymmetric m-adic, which changes sign under the transposition of any two "vector locations" (in index notation a tensor of order m which changes sign under the interchange of any two indices).

[23] It should be noted that $d\mathbf{x}\, d\mathbf{x}$ (often written $d\mathbf{x} \wedge d\mathbf{x}$ in the literature) is, despite appearances, neither symmetric nor a dyad, but rather an antisymmetric dyadic. Indeed, since $\mathfrak{A}$ is antisymmetric, any symmetric component of $d\mathbf{x}\, d\mathbf{x}$ would contribute nothing to the differential form. The interchange of vector order reverses the orientation of the surface element $d\mathbf{x}\, d\mathbf{x}$ and with it the sign. In fact, on any surface determined by giving $\mathbf{x}$ as a function of two parameters u and v, we have $d\mathbf{x}\, d\mathbf{x} = (\mathbf{x}_u\mathbf{x}_v - \mathbf{x}_v\mathbf{x}_u)\, du\, dv$.

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it is taken over. This is the case only if $\mathbf{C}^*$ is a gradient (i.e., a $\mathbf{z}$ derivative), so that its curl vanishes,

$$\mathbf{C}^*_{\mathbf{z}} - {}_{\mathbf{z}}\mathbf{C}^* = 0. \tag{D13}$$

But then we are amusingly compensated in that the phase integral of (D10) can be written

$$\mathfrak{B}^*_s + \mathfrak{B}^* \cdot \epsilon \mathbf{h}_{\mathbf{z}} + \epsilon\, {}_{\mathbf{z}}\mathbf{h} \cdot \mathfrak{B}^* = (\mathbf{W}^* + \mathbf{C}^*\,\omega)_{\mathbf{z}} - {}_{\mathbf{z}}(\mathbf{W}^* + \mathbf{C}^*\,\omega), \tag{D14}$$

which is just the condition [analogous to (D7)] that $\mathfrak{B}^* : d\mathbf{z}\, d\mathbf{z}$ be a relative integral invariant of the reduced system.

7. Inheritance of Absolute Invariants

There was an element of humbug in the just described inheritance of relative invariants. The fact is that any relative invariant of a system yields lower order relative invariants upon integration over closed sub-manifolds, so that if the system is splittable, then when the nonautonomous variables are integrated out, a relative invariant of the new autonomous subsystem is obtained. That the "niceness" of the nice variables was really irrelevant is apparent in that the theorem of phase independence was not invoked. With the inheritance of absolute invariants the situation is otherwise, and we are back in our subject proper.

If $\mathfrak{A} : d\mathbf{x}\, d\mathbf{x}$ is an absolute invariant, then transforming to nice variables and introducing $\mathfrak{B}$ and $\mathbf{C}$ by (D9) as before, we have (D10) and (D11) with the right-hand sides replaced by zero. From the latter we see, first, by the theorem of phase independence, that $\mathbf{C}$ depends only on $\mathbf{z}$, and second, as a consequence, that $\mathbf{C} \cdot d\mathbf{z}$ is an absolute invariant of the reduced system. From the former, we then see that $\mathfrak{B}$ depends only on $\mathbf{z}$, but it is not apparent what more can be deduced, unless $\mathbf{C} \cdot d\mathbf{z}$ is a trivial invariant, in which case $\mathfrak{B} : d\mathbf{z}\, d\mathbf{z}$ is easily seen as before to be a relative invariant (or unless either of the very special cases $\mathbf{C} = 0$ or $\omega_{\mathbf{z}} = 0$ obtains, when $\mathfrak{B} : d\mathbf{z}\, d\mathbf{z}$ is of course even an absolute invariant).

E. HAMILTONIAN SYSTEMS

1. Preparatory Transformations

The hereditary property with the perhaps most important applications is that of being in Hamiltonian form. Let the dependent variables consist of the $N = 2M$ canonically conjugate coordinates $\mathbf{q}$ and $\mathbf{p}$, and let the Hamiltonian $H(\mathbf{q}, \mathbf{p})$ be single-valued and time-independent. (Time-dependent systems, such as those with "slowly changing external parameters," are not excluded by this, since they can be converted to time-independent form by the well-known device of treating time and energy as a new additional pair of conjugate coordinates. See also Sec. B.4.) The $\mathbf{q}$ and $\mathbf{p}$ jointly satisfy the autonomous system constituted by Hamilton's equations

$$\mathbf{q}_t = H_{\mathbf{p}}, \qquad \mathbf{p}_t = -H_{\mathbf{q}}. \tag{E1}$$

In order to allow the widest latitude in applications, we do not require that (E1) be in standard form, but merely that there exist some (monotonic) transformation of the independent variable

$$t \leftrightarrow s \tag{E2}$$

and some transformation of the dependent variables

$$\mathbf{q}, \mathbf{p} \leftrightarrow \mathbf{x} \tag{E3}$$

which take the system into the standard form (B4) with all solutions nearly periodic. (The only reason for permitting the former transformation is to normalize the magnitude of $\mathbf{f}$; therefore, t and s will generally differ at most by a factor which is some power of ϵ.) Applying now our general theory, we make a further transformation of dependent variables

$$\mathbf{q}, \mathbf{p} \leftrightarrow \mathbf{x} \leftrightarrow \mathbf{z}, \phi \tag{E4}$$

to nice variables satisfying (B18). Here $\mathbf{x}$ has been a mere intermediary and is of no further interest.

2. Action Integral

It is well known that $\mathbf{p} \cdot d\mathbf{q}$ is a relative integral invariant (see Sec. D.5) of every Hamiltonian system; indeed, this property may be used to define Hamiltonian systems. In other words, the line integral $\oint \mathbf{p} \cdot d\mathbf{q}$ taken around any closed curve remains constant in the course of time if (every point of) the curve varies in accordance with the equations of motion (E1). In this way one may construct any number of constants of motion, which, however, are for the most part not useful, because not locally computable (see Sec. A.4). There is no way of telling what the family of closed curves chosen at one time will have become at a later time, except of course by fully solving the equations of motion, the very task to avoid or (to help) to effect which one would like to know constants of motion in the first place.

In the present situation there is, nevertheless, a family of curves admirably suited to the construction of a useful constant of motion; these are the

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rings, which as a family remain invariant (since rings flow into rings; see Sec. C.3), and which are to an adequate degree locally computable (not local to a point, but local to the loop through a point). Accordingly we define the "phase integral of action"

$$J(\mathbf{z}) \equiv \oint_{\text{ring } \mathbf{z}} \mathbf{p}\cdot d\mathbf{q} = \int_0^1 d\phi\, \mathbf{p}\cdot\mathbf{q}_\phi, \qquad \text{(E5)}$$

in which $\mathbf{q}$ and $\mathbf{p}$ are to be thought of as functions of $\mathbf{z}$ and ϕ in accordance with (E4). It is perhaps worth giving the trivial direct proof of the constancy of J to all orders: since differentiation with respect to s commutes with differentiation or integration with respect to ϕ, so does differentiation with respect to t, whence

$$J_t = \oint (\mathbf{p}_t\cdot d\mathbf{q} - \mathbf{q}_t\cdot d\mathbf{p})$$

$$= -\oint (H_\mathbf{q}\cdot d\mathbf{q} + H_\mathbf{p}\cdot d\mathbf{p}) = -\oint dH = 0, \qquad \text{(E6)}$$

where in the first step we have integrated the second term by parts and in the next we have used (E1). (Incidentally, J is obviously also invariant under canonical transformations.)

The constancy of J to all orders clearly holds for ranges of the independent variable of order ϵ^{-n} for any n, so long as J remains defined.

3. Poisson Brackets

Let $\mathcal{A}$ and $\mathcal{B}$ be any polyadic functions of the state of our physical system, i.e., of $\mathbf{q}$, $\mathbf{p}$. Their *Poisson bracket* is defined (as standardly) by

$$[\mathcal{A}, \mathcal{B}] = \mathcal{A}_\mathbf{q}\cdot_\mathbf{p}\mathcal{B} - \mathcal{A}_\mathbf{p}\cdot_\mathbf{q}\mathcal{B}. \qquad \text{(E7)}$$

We shall need the formula

$$[\mathcal{A}, \mathcal{B}]_s = [\mathcal{A}_s, \mathcal{B}] + [\mathcal{A}, \mathcal{B}_s], \qquad \text{(E8)}$$

which is immediately equivalent to its analogue with t derivatives in place of s derivatives (multiply by t_s or s_t to go back or forth), a well known formula easily derived by straightforward calculation using nothing more than (E1).

Now, by (E8), (B18), and (E7) we have

$$[\mathbf{z}, \mathbf{z}]_s = [\epsilon\,\mathbf{h}, \mathbf{z}] + [\mathbf{z}, \epsilon\,\mathbf{h}]$$

$$= \epsilon\,\mathbf{h}_\mathbf{z}\cdot[\mathbf{z}, \mathbf{z}] + \epsilon\,[\mathbf{z}, \mathbf{z}]\cdot_\mathbf{z}\mathbf{h}, \qquad \text{(E9)}$$

so that by the theorem of phase independence the (obviously antisymmetric) dyadic $[\mathbf{z}, \mathbf{z}]$ is a function of $\mathbf{z}$ alone,

$$[\mathbf{z}, \mathbf{z}]_\phi = 0. \qquad \text{(E10)}$$

{Note that although there is no reason to suppose that $[\mathbf{z}, \mathbf{z}]$ is $O(1)$, the theorem is applicable anyway because (E9) is linear.} In the same way we have

$$[\phi, \mathbf{z}]_s = [\omega, \mathbf{z}] + [\phi, \epsilon\,\mathbf{h}]$$

$$= \omega_\mathbf{z}\cdot[\mathbf{z}, \mathbf{z}] + \epsilon\,[\phi, \mathbf{z}]\cdot_\mathbf{z}\mathbf{h}, \qquad \text{(E11)}$$

so that, again by the theorem of phase independence, which applies in view of (E10),

$$[\phi, \mathbf{z}]_\phi = 0. \qquad \text{(E12)}$$

We need the two formulas

$$[\mathbf{z}, \mathbf{z}]\cdot({}_\mathbf{z}\mathbf{p}\cdot\mathbf{q}_\phi - {}_\mathbf{z}\mathbf{q}\cdot\mathbf{p}_\phi) = 0, \qquad \text{(E13)}$$

$$[\phi, \mathbf{z}]\cdot({}_\mathbf{z}\mathbf{p}\cdot\mathbf{q}_\phi - {}_\mathbf{z}\mathbf{q}\cdot\mathbf{p}_\phi) = 1. \qquad \text{(E14)}$$

The short way to establish them is to observe that their common parenthetical factor is the Lagrange bracket of $\mathbf{z}$ and ϕ and to exploit the well-known relationship between the Poisson and the Lagrange brackets, in view of the fact that the summation indicated by the left-most dot in each of them runs over a set of variables complete except for the omission of ϕ, for which the Lagrange bracket would vanish anyway. Of course (E13) and (E14) can also be easily established by direct calculation. It is only necessary to expand them out by (E7) and the distributive law and then to make use of relations (coming from the chain rule for differentiation) between the derivatives of $\mathbf{q}$ and $\mathbf{p}$ with respect to $\mathbf{z}$ and ϕ and *vice versa*, namely

$$\mathbf{q}_\mathbf{z}\cdot\mathbf{z}_\mathbf{q} + \mathbf{q}_\phi\,\phi_\mathbf{q} = \mathbf{q}_\mathbf{q} = \mathcal{I} \qquad \text{(E15)}$$

and similar relations based on two-sided evaluations of $\mathbf{q}_\mathbf{p}$, $\mathbf{p}_\mathbf{q}$, $\mathbf{p}_\mathbf{p}$, $\mathbf{z}_\phi$, and ϕ_ϕ. (The two remaining similar relations coming from $\mathbf{z}_\mathbf{z}$ and $\phi_\mathbf{z}$ are not needed.)

We are now prepared to evaluate two particular Poisson brackets of interest. One is

$$[\mathbf{z}, J] = [\mathbf{z}, \mathbf{z}]\cdot J_\mathbf{z}$$

$$= [\mathbf{z}, \mathbf{z}]\cdot\int_0^1 d\phi\,({}_\mathbf{z}\mathbf{p}\cdot\mathbf{q}_\phi - {}_\mathbf{z}\mathbf{q}\cdot\mathbf{p}_\phi) = 0, \qquad \text{(E16)}$$

where the first equation is seen to hold by (E7) and the fact that J depends on $\mathbf{z}$ alone, the second by direct differentiation of (E5) followed by an integration by parts, and the last by (E13) after $[\mathbf{z}, \mathbf{z}]$ has been brought inside the integral by virtue of (E10). The other is

$$[\phi, J] = 1, \qquad \text{(E17)}$$

as seen in the same way using (E14) and (E12).

4. Nontriviality of Action Integral

We digress momentarily to point out that J can hardly be a trivial constant of motion. It certainly cannot be completely trivial, in the sense of being an outright constant, for then its Poisson bracket with any other quantity would vanish,

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contradicting (E17). But suppose that J can be expressed as a function of H alone, which is obviously a kind of triviality since H is, of course, automatically a constant of motion. Now J_H cannot vanish, else we would have $[\phi, J] = [\phi, H]J_H = 0$, contradicting (E17) as before, so we may invert the relationship and express H as a function of J. Then

$$\mathbf{z}_t = [\mathbf{z}, H] = [\mathbf{z}, J]H_J = 0, \qquad \text{(E18)}$$

where the first equation is well known and obvious, and the last is a consequence of (E16). Thus the $\mathbf{z}$ provide a complete set of integrals of the system. Only in this extremely special case can J be a function of H alone.

5. Reduced Hamiltonian System

We continue with our program of showing the hereditary character of being Hamiltonian. To this end we wish to put the reduced system into Hamiltonian form. Through the reduction process itself we have gotten rid of (or "left behind") one variable, ϕ, and now that we have the nontrivial constant of motion J, we may use it to eliminate another variable. Thus we may hope to put the remaining system of $2M - 2$ variables into the form of a Hamiltonian system of $M - 1$ degrees of freedom.

Our actual procedure for obtaining this *reduced time-independent Hamiltonian* system is to make a canonical transformation from our original set of M pairs of canonically conjugate coordinates $(\mathbf{q}, \mathbf{p})$ to a new set of which one pair is (ϕ, J) and of which the remaining $M - 1$ pairs, to be denoted collectively by $(\mathbf{Q}, \mathbf{P})$, are also locally computable. Now the necessary and sufficient condition that the transformation to a new set of pairs of coordinates be canonical is that the Poisson bracket of any two new coordinates be zero if they come from different pairs and unity if they constitute a pair (in the right order). Because of (E17) it is not immediately excluded that we can find a canonical transformation of the kind described, but it is also not immediately obvious that we can. In Appendix 2 it is shown[24] that whenever one knows some of the new coordinates of a proposed prospective canonical transformation, and these coordinates satisfy all the requisite Poisson bracket conditions among themselves and have linearly independent derivatives with respect to the old coordinates, then one can indeed define the missing coordinates (in a way locally dependent on the Hamiltonian) so as to form a complete canonical transformation; furthermore the whole construction is obviously valid order by order when the coordinates involved are construed as series.

The condition of linear independence of the derivatives is certainly satisfied, for if we had, say,

$$J_{\mathbf{q}} = k\,\phi_{\mathbf{q}}, \qquad J_{\mathbf{p}} = k\,\phi_{\mathbf{p}}, \qquad \text{(E19)}$$

we could immediately deduce

$$[\phi, J] = k[\phi, \phi] = 0,$$

once again contradicting (E17). Accordingly, we may assume that we now have the sequence of transformations

$$\phi, \mathbf{z} \leftrightarrow \mathbf{q}, \mathbf{p} \leftrightarrow \Phi, J, \mathbf{Q}, \mathbf{P}, \qquad \text{(E20)}$$

where, in view of the discussion in Sec. A.3, we have introduced the new symbol Φ (not to be confused with the Φ of part B) for ϕ in its role as a member of the new set of variables, so that

$$\Phi = \phi \qquad \text{(E21)}$$

may be used freely except to substitute one subscript for another (but see below!). The new variables $\mathbf{Q}$ and $\mathbf{P}$ have been so constructed as to satisfy the correct bracket conditions

$$[\mathbf{Q}, \mathbf{Q}] = 0, \quad [\mathbf{Q}, \mathbf{P}] = \mathcal{I}, \quad [\mathbf{P}, \mathbf{P}] = 0 \qquad \text{(E22)}$$

among themselves and also the "cross-conditions"

$$0 = [\mathbf{Q}, \Phi] = [\mathbf{Q}, \phi] = \mathbf{Q}_{\mathbf{z}} \cdot [\mathbf{z}, \phi], \qquad \text{(E23)}$$

$$0 = [\mathbf{Q}, J] = \mathbf{Q}_{\phi}\,[\phi, J] + \mathbf{Q}_{\mathbf{z}} \cdot [\mathbf{z}, J] = \mathbf{Q}_{\phi}, \qquad \text{(E24)}$$

and the same with $\mathbf{Q}$ replaced throughout by $\mathbf{P}$. From (E24) we see that $\mathbf{Q}$ and $\mathbf{P}$ actually are independent of ϕ, like J, so that the transformation $\mathbf{z} \leftrightarrow J, \mathbf{Q}, \mathbf{P}$ is wholly independent of ϕ, and we may (and from now on do) obliterate completely the distinction between Φ and ϕ.

The new Hamiltonian is of course just the old one H itself, naturally expressed now as a function of the new canonical coordinates through the transformation (E20). In the new coordinates, Hamilton's equations take the form

$$\phi_t = H_J. \qquad \text{(E25)}$$

[24] A very similar theorem is demonstrated by Nordheim and Fues [L. Nordheim and E. Fues, *Handbuch der Physik*, edited by S. Flügge (Julius Springer, Berlin, 1927) B.5, Kap. 3, Ziff. 10], who assume that an integral is known and show how this can be used to reduce to a Hamiltonian system of one less degree of freedom; one finds a canonical transformation to new coordinates one of which is the known integral, in which case its conjugate coordinate obviously is ignorable. In our present application the ignorable coordinate ϕ has come first. It is not on account of this minor difference that a separate discussion is appended, but rather partly to keep this paper as self-contained as possible and partly because the method is so simple and seems to offer a slightly unconventional insight into the nature of the Poisson bracket conditions.

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$$J_t = -H_\phi, \tag{E26}$$

$$\mathbf{Q}_t = H_\mathbf{P}, \qquad \mathbf{P}_t = -H_\mathbf{Q}. \tag{E27}$$

Since $\phi_t = \phi_s s_t = \omega s_t$, (E25) yields the familiar sort of relation between the frequency and the derivative of the Hamiltonian with respect to the action variable. Since $J_t = 0$, (E26) shows that H is independent of ϕ, which we might have known anyway from the fact that H is an integral of the system (see Sec. D.3). It is now clear that (E27) constitutes a one-parameter family (parametrized by J) of autonomous Hamiltonian systems.

6. Possibility of Repetition and Further Invariants

As a consequence of our ability to put the reduced system back into Hamiltonian form, we see immediately that if the reduced system itself has all solutions nearly periodic (naturally with a period longer than the period of the original system at least by an order of ϵ), then there exists also another constant of motion

$$J' \equiv \oint_{\text{ring } \mathbf{z}'} \mathbf{P}\cdot d\mathbf{Q} = \int_0^1 d\phi'\, \mathbf{P}\cdot\mathbf{Q}_{\phi'}, \tag{E28}$$

where $\mathbf{z}'$ and ϕ' are the new nice variables for the reduced system,

$$\mathbf{z} \leftrightarrow J, \mathbf{Q}, \mathbf{P} \leftrightarrow \mathbf{z}', \phi' \tag{E29}$$

As one should expect, it is by no means necessary to go through the complicated integration process involved in finding $\mathbf{Q}$ and $\mathbf{P}$ in order to compute J', for

$$J' = \int_0^1 d\phi' \,(\mathbf{P}\cdot\mathbf{Q}_{\phi'} + J\,\phi_{\phi'}) = \int_0^1 d\phi'\, \mathbf{p}\cdot\mathbf{q}_{\phi'}, \tag{E30}$$

since $\phi_{\phi'} = 0$ [the transformation (E29) being independent of ϕ] and the differential form $\mathbf{p}\cdot d\mathbf{q}$ is a canonical invariant.

Because $\mathbf{Q}$ and $\mathbf{P}$ are locally computable, so is J'. Furthermore, J' cannot depend on J only, nor even on J and H only except in the very trivial special case that we can find a complete set of integrals. Incidentally it may be noted that there is no chance here of finding still another integral by the well-known device of forming the Poisson bracket of two already known, since by a further canonical transformation we can introduce a set of canonical coordinates among which are the conjugate pairs (ϕ, J) and (ϕ', J'), which demonstrates that

$$[J, J'] = 0 \tag{E31}$$

in addition to three other such relations.

If the new reduced system should itself happen to have all solutions nearly periodic we can find still another integral J'' and ignore another angle variable ϕ'', and so on indefinitely, till the successive near periodicities run out. If the internal degrees of freedom run out first, we end up with a complete (asymptotic) description of the system in terms of M ignorable angle variables and their M locally determined conjugate action variables.

F. APPENDICES

1. Fundamental Approximation Theorem

For the sake of completeness we give here a brief outline of the well-known proof that two functions which satisfy nearly identical autonomous systems and have nearly equal initial values are themselves nearly equal for a bounded range of the independent variable. Let $\mathbf{x}(s)$ and $\mathbf{x}^\dagger(s)$ satisfy

$$\mathbf{x}_s = \mathbf{f}(\mathbf{x}), \qquad \mathbf{x}^\dagger_s = \mathbf{f}(\mathbf{x}^\dagger) + \boldsymbol{\gamma}(\mathbf{x}^\dagger), \tag{F1}$$

$$\mathbf{x}(0) = \mathbf{x}^\dagger(0) + \boldsymbol{\delta}, \tag{F2}$$

where $\boldsymbol{\gamma}$ and $\boldsymbol{\delta}$ are small quantities whose norms (maximized over $\mathbf{x}$ space in the case of $\boldsymbol{\gamma}$) we denote by γ and δ. (Any reasonable norm may be employed, and will be denoted by absolute value signs.) In the region of interest in $\mathbf{x}$ space $\mathbf{f}$ satisfies a Lipschitz condition,

$$|\mathbf{f}(\mathbf{x}') - \mathbf{f}(\mathbf{x}'')| \le c\,|\mathbf{x}' - \mathbf{x}''| \tag{F3}$$

for an appropriate positive constant c and any $\mathbf{x}'$ and $\mathbf{x}''$ (if only because $\mathbf{f}_\mathbf{x}$ exists and is continuous on a closed bounded domain and therefore has all components bounded). To estimate the difference we have

$$\begin{aligned} |\mathbf{x}^\dagger - \mathbf{x}| &= \Big|\mathbf{x}^\dagger(0) - \mathbf{x}(0) \\ &\quad + \int_0^s ds\,[\mathbf{f}(\mathbf{x}^\dagger) + \boldsymbol{\gamma}(\mathbf{x}^\dagger) - \mathbf{f}(\mathbf{x})]\Big| \\ &\le \delta + \int_0^s ds\,[c\,|\mathbf{x}^\dagger - \mathbf{x}| + \gamma], \end{aligned} \tag{F4}$$

for simplicity restricting ourselves to non-negative values of s. Now the maximum value $\mu(s)$ of $|\mathbf{x}^\dagger - \mathbf{x}|$ in the range from 0 to s satisfies

$$\begin{aligned} \mu &\le \delta + \int_0^s ds\,[c\mu + \gamma] \\ &\le \delta + s[c\mu + \gamma], \end{aligned} \tag{F5}$$

so that for $s \le 1/2c$ (for instance) we have

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$$\mu \leq (\delta + \gamma s)/(1 - cs) \leq 2\delta + 2\gamma s. \tag{F6}$$

In order to obtain an estimate for larger values of s we could apply (F6) over and over again, proceeding in steps of size $1/2c$ and using the final estimate at each step as the initial estimate corresponding to (F2) for the next step. In this way we would obtain an exponential estimate, which is, however, more elegantly and perhaps simply obtained by noting that an upper bound for $|\mathbf{x}^\dagger - \mathbf{x}|$ is provided by the function $\nu(s)$ which satisfies the equation obtained from (F4) by replacing $|\mathbf{x}^\dagger - \mathbf{x}|$ by ν and $\leq$ by $=$. Indeed, the difference $D \equiv \nu - |\mathbf{x}^\dagger - \mathbf{x}|$ satisfies

$$D \geq c \int_0^s ds\, D, \tag{F7}$$

$$\left(e^{-cs} \int_0^s ds\, D\right)_s \geq 0, \tag{F8}$$

whence upon integration from 0 to s it follows that the right-hand side of (F7) and *a fortiori* D are nonnegative. But it is trivial to solve explicitly for ν, and we find that

$$|\mathbf{x}^\dagger - \mathbf{x}| \leq \nu = (\delta + \gamma/c)e^{cs} - \gamma/c. \tag{F9}$$

That there can be no estimate radically better than this is obvious from even the simplest examples.

The standard uniqueness theorem is of course obtained as the special case $\boldsymbol{\gamma} = \boldsymbol{\delta} = 0$.

2. Completion of Canonical Transformation

The necessary and sufficient conditions that a transformation

$$\mathbf{q}, \mathbf{p} \leftrightarrow \mathbf{q}^\dagger, \mathbf{p}^\dagger \tag{F10}$$

be canonical are of course the fundamental Poisson bracket relations

$$[\mathbf{q}^\dagger, \mathbf{q}^\dagger] = 0, \quad [\mathbf{q}^\dagger, \mathbf{p}^\dagger] = \mathcal{I}, \quad [\mathbf{p}^\dagger, \mathbf{p}^\dagger] = 0. \tag{F11}$$

A consequence of these relations is that the derivatives of the components of $\mathbf{q}^\dagger$ and $\mathbf{p}^\dagger$ are linearly independent (and *a fortiori* that the components themselves are functionally independent), since for any linear combination of these derivatives which vanishes,

$$\mathbf{a}\cdot\mathbf{q}_\mathbf{q}^\dagger + \mathbf{b}\cdot\mathbf{p}_\mathbf{q}^\dagger = 0, \qquad \mathbf{a}\cdot\mathbf{q}_\mathbf{p}^\dagger + \mathbf{b}\cdot\mathbf{p}_\mathbf{p}^\dagger = 0, \tag{F12}$$

we have, by (F11) and (F12), that

$$\mathbf{a} = \mathbf{a}\cdot[\mathbf{q}^\dagger, \mathbf{p}^\dagger] = -\mathbf{b}\cdot[\mathbf{p}^\dagger, \mathbf{p}^\dagger] = 0 \tag{F13}$$

and similarly that $\mathbf{b} = 0$.

Suppose now that we are given, as functions of $\mathbf{q}$ and $\mathbf{p}$, some but not all of the components of $\mathbf{q}^\dagger$ and $\mathbf{p}^\dagger$ in a proposed transformation (F10). We wish to prove that we can construct the remaining components. Let $\mathbf{q}$, $\mathbf{p}$, $\mathbf{q}^\dagger$, $\mathbf{p}^\dagger$ each have M components. We denote the $N < 2M$ given components (of both $\mathbf{q}^\dagger$ and $\mathbf{p}^\dagger$, not necessarily paired) collectively by $\boldsymbol{\alpha}$, and require them to have linearly independent derivatives and to satisfy those components of relations (F11) which involve them alone. Since we can proceed by mathematical induction, it suffices to construct one of the remaining components of $\mathbf{q}^\dagger$ or $\mathbf{p}^\dagger$, say of $\mathbf{q}^\dagger$. We denote this component by q' and its conjugate by p'. We must construct q' to satisfy those components of relations (F11) which involve it and the $\boldsymbol{\alpha}$. These conditions on q' have in fact the form

$$[\boldsymbol{\alpha}, q'] = \mathbf{c}, \tag{F14}$$

where $\mathbf{c}$ is constant and, to be sure, has all components zero except for at most one of them, which equals -1 (that component, if it is among the $\boldsymbol{\alpha}$, which corresponds to p').

Conditions (F14) amount to a prescription of the directional derivative of q' in each of a number of directions, one for each component of $\boldsymbol{\alpha}$. These directions are linearly independent, since the derivatives of the components of $\boldsymbol{\alpha}$ are. We next inquire into the conditions for the compatibility of (F14), namely the conditions that the second derivatives of q' be the same independently of the order of differentiation. The dyadic second directional derivative of q' (which, if expanded out, contains first as well as second derivatives of q') is

$$[\boldsymbol{\alpha}, [\boldsymbol{\alpha}, q']] = [\boldsymbol{\alpha}, \mathbf{c}] = 0, \tag{F15}$$

since $\mathbf{c}$ is constant. Taking the directional derivatives in the reverse order gives

$$-[[\boldsymbol{\alpha}, q'], \boldsymbol{\alpha}] = 0 \tag{F16}$$

[which happens to be the transpose of (F15)]. This has the same second derivatives of q', so we obtain the consistency conditions by taking the difference

$$[\boldsymbol{\alpha}, [\boldsymbol{\alpha}, q']] + [[\boldsymbol{\alpha}, q'], \boldsymbol{\alpha}] = 0. \tag{F17}$$

Now, Jacobi's identity, written for two vectors and a scalar with due regard to the dyadic order, is

$$[\boldsymbol{\alpha}, [\boldsymbol{\alpha}', q']] + [[\boldsymbol{\alpha}, q'], \boldsymbol{\alpha}'] + [q', [\boldsymbol{\alpha}, \boldsymbol{\alpha}']] = 0, \tag{F18}$$

the middle term being the transpose of $[\boldsymbol{\alpha}', [q', \boldsymbol{\alpha}]]$ Taking $\boldsymbol{\alpha}'$ to be $\boldsymbol{\alpha}$, we see that (F17) is equivalent to

$$[[\boldsymbol{\alpha}, \boldsymbol{\alpha}'], q'] = 0, \tag{F19}$$

which is explicitly devoid of second derivatives of

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q'. But $[\alpha, \alpha']$ is a constant dyadic, all of whose components are 0 or ± 1, in view of (F11), so (F19) holds identically.

Since the compatibility conditions are satisfied (tautologously), it follows from the general theory of partial differential equations that (F14) has a general solution q' with the freedom of an arbitrary function of $2M - N$ variables. From the geometrical point of view, indeed, we have shown that the N-dimensional vector space determined at each point by the directions of the directional derivatives $[\alpha, \cdots]$ there, is the tangent space of a $(2M - N)$-dimensional family of N-dimensional hypersurfaces.[25] In (F14) we have a complete prescription of the derivatives of q' as a function on each hypersurface, but no restriction on the variation of q' from one hypersurface to another. Clearly q' is determined up to a constant (its "initial value" at any given point) on each hypersurface.

It remains only to show that the arbitrariness in q' permits us to choose it in such a way that the derivatives of q' and the α will be linearly independent. This is merely a matter of counting dimensions, for suppose contrarily that

$$q'_{\mathfrak{q}} = \mathbf{k}\cdot\boldsymbol{\alpha}_{\mathfrak{q}}, \qquad q'_{\mathfrak{p}} = \mathbf{k}\cdot\boldsymbol{\alpha}_{\mathfrak{p}}. \tag{F20}$$

If p' is among the α, then since $[\alpha, q']$ vanishes in all except that component, and since by (F20)

$$0 = [q', q'] = \mathbf{k}\cdot[\alpha, q'], \tag{F21}$$

it follows that the corresponding component of $\mathbf{k}$ vanishes. Similarly, if any (other) two conjugate coordinates q'', p'' are among the α, the corresponding components of $\mathbf{k}$ must both vanish, else by means of (F20) we could solve for and eliminate the derivatives of either q'' or p'' and by (F11) obtain the contradiction

$$1 = [q'', p''] = 0.$$

Let the number of such pairs be L, so that the number of unpaired components of α is $N - 2L$. Obviously

$$N - L = (N - 2L) + L \leq M, \tag{F22}$$

the right-hand equality holding only if p' is among the α. The number of possibly nonzero components of $\mathbf{k}$ is

$$N - 2L = 2(N - L) - N \leq 2M - N \tag{F23}$$

or $N - 2L - 1$ if p' is among the α; thus in either case it is strictly less than $2M - N$. But q' has the freedom of an arbitrary function of $2M - N$ variables, so its derivatives span a $(2M - N)$-dimensional linear space, contradicting (F20).

3. Iterated Near-Identity Mappings

There is a strong and intimate connection between the asymptotic theory of nearly recurrent systems and the asymptotic theory of iterated near-identity mappings, which has previously been treated less satisfactorily by Kruskal[5] and applied by Spitzer.[4] In this section we describe the latter theory and explore the connections.

A *near-identity mapping* of a space of vectors ξ into itself is a function, $\mathbf{T}(\xi)$ say, which can be written in the form

$$\mathbf{T}(\xi) = \xi + \epsilon\,\mathbf{D}(\xi), \tag{F24}$$

where $\mathbf{D}(\xi)$ may be a series in powers of ϵ and is required to be $O(1)$. Associated with such a transformation (as with any) is the sequence of its iterates, the transformations $\mathbf{T}^n(\xi)$ defined recursively by

$$\mathbf{T}^1(\xi) \equiv \mathbf{T}(\xi), \qquad \mathbf{T}^{n+1}(\xi) \equiv \mathbf{T}^n(\mathbf{T}(\xi)), \tag{F25}$$

where n runs over the positive integers.[26] We then ask for an asymptotic description (as $\epsilon \to 0$) of the discrete "trajectory" of a point ξ under iteration of $\mathbf{T}$, i.e., of the sequence of points $\mathbf{T}^n(\xi)$, $n = 0, 1, 2, \cdots$.

Obviously as $\epsilon \to 0$ the points get closer and closer together. For any fixed n, of course $\mathbf{T}^n(\xi) \to \xi$, but this does not imply that the trajectory as a whole collapses into the initial point, since for n taken approximately equal to c/ϵ (where c is a constant) we may expect $\mathbf{T}^n(\xi)$ to remain distinctly away from ξ and indeed to converge to a limit[27] different from ξ. Rather, the discrete trajectory approximates better and better to a continuous (and smooth) curve denoted by $\boldsymbol{\Xi}^{(0)}(\sigma)$ and determined by

$$\boldsymbol{\Xi}^{(0)}(0) = \xi, \qquad \boldsymbol{\Xi}^{(0)}_{\sigma} = \mathbf{D}(\boldsymbol{\Xi}^{(0)}), \tag{F26}$$

the latter being obtained from (F24) by replacing ξ by $\mathbf{T}^n(\xi)$ and then going to the limit after making the identifications

$$\sigma \equiv n\epsilon, \tag{F27}$$

[25] See Newcomb [W. Newcomb, Ann. Phys. **3**, 347 (1958)] for a heuristic description of the theory of such questions. In his terminology, we have shown that the "cross-bracket" of any two directions of differentiation is zero.

[26] Obviously (F24) can be formally inverted to yield an inverse transformation $\mathbf{T}^{-1}$, with iterates $\mathbf{T}^{-2}$, $\mathbf{T}^{-3}$, etc. With $\mathbf{T}^0(\xi) \equiv \xi$, (F25) becomes valid for all integers n.

[27] This limit can in fact be expressed as $\boldsymbol{\Xi}^{(0)}(c)$, see (F28).

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$$\mathbf{\Xi}^{(0)}(\sigma) \equiv \lim_{\epsilon=0} \mathbf{T}^n(\xi). \tag{F28}$$

Except to zeroth order the points $\mathbf{T}^n(\xi)$ naturally need not lie on $\mathbf{\Xi}^{(0)}$. This suggests (what we have already presaged by the notation) that we seek a formal series $\mathbf{\Xi}(\sigma)$ such that the representation

$$\mathbf{T}^n(\xi) = \mathbf{\Xi}(\sigma) \tag{F29}$$

will be valid to all orders under the identification (F27), which there is no need to modify. And it is indeed easy to obtain the appropriate conditions (differential equation and initial condition) determining $\mathbf{\Xi}$ to higher orders. We could then develop a rigorous theory, with analogues of the results in the main body of this paper. It is simpler, however, to reduce the problem to the one that we have already exhaustively analyzed.

The crucial step is to imbed the sequence $\mathbf{T}^n(\xi)$ in a continuous trajectory satisfying an autonomous system of differential equations. That is, we seek a function $\mathbf{n}(\xi)$ such that the curve $\xi^\dagger(\xi, \sigma)$ defined as the solution of the system

$$\xi^\dagger_\sigma = \mathbf{n}(\xi^\dagger) \tag{F30}$$

determined by the initial condition

$$\xi^\dagger(\xi, 0) = \xi \tag{F31}$$

passes through the points in the sequence for σ given by (F27):

$$\xi^\dagger(\xi, n\epsilon) = \mathbf{T}^n(\xi). \tag{F32}$$

We shall show that $\mathbf{n}$ is determined (and uniquely) as a series to all orders by this condition (required to hold for arbitrary ξ).

Obviously it suffices to determine $\mathbf{n}$ merely so that (F32) holds with $n = 1$,

$$\xi^\dagger(\xi, \epsilon) = \mathbf{T}(\xi), \tag{F33}$$

since it will then automatically hold for all larger n by an induction based on the calculation

$$\begin{aligned}\xi^\dagger(\xi, n\epsilon + \epsilon) &= \xi^\dagger(\xi^\dagger(\xi, \epsilon), n\epsilon) \\ &= \xi^\dagger(\mathbf{T}(\xi), n\epsilon) = \mathbf{T}^n(\mathbf{T}(\xi)) = \mathbf{T}^{n+1}(\xi),\end{aligned} \tag{F34}$$

where the first equation is an instance of (A7), the second will follow from (F33), the third will be an instance of the induction hypothesis (F32), and the fourth is part of (F25). Now by a Taylor expansion

$$\begin{aligned}\xi^\dagger(\xi, \epsilon) &= \xi^\dagger(\xi, 0) + \epsilon\, \xi^\dagger_\sigma(\xi, 0) + \tfrac{1}{2}\epsilon^2\, \xi^\dagger_{\sigma\sigma}(\xi, 0) + \cdots \\ &= \xi + \epsilon\, \mathbf{n}(\xi) + \tfrac{1}{2}\epsilon^2\, \mathbf{n}_\xi(\xi)\cdot\mathbf{n}(\xi) + \cdots,\end{aligned} \tag{F35}$$

which by (F24) and (F33) may be written in the form

$$\mathbf{n} + \tfrac{1}{2}\epsilon\, \mathbf{n}_\xi\cdot\mathbf{n} + \cdots = \mathbf{D}, \tag{F36}$$

omitting the argument ξ everywhere. Inverting this we have immediately

$$\mathbf{n} = \mathbf{D} - \tfrac{1}{2}\epsilon\, \mathbf{D}_\xi\cdot\mathbf{D} + \cdots. \tag{F37}$$

The desired representation (F29) is now obviously obtained merely by taking

$$\mathbf{\Xi}(\sigma) \equiv \xi^\dagger(\xi, \sigma). \tag{F38}$$

It is at last a trivial matter to bring this theory under that of the main body of the paper. Treating n as a continuous variable and calling it s now, or in other words defining

$$s \equiv \sigma/\epsilon, \tag{F39}$$

and introducing a redundant angle variable θ equal to s up to an arbitrary additive integer, and finally defining[28]

$$\mathbf{x} \equiv (\xi, \theta), \tag{F40}$$

we have a system in standard form

$$\mathbf{x}_s = \mathbf{f} \equiv (\epsilon\, \mathbf{n}, 1). \tag{F41}$$

This is nearly recurrent, since to zeroth order ξ is constant and θ increases uniformly, returning to its initial value (up to an integer, which is all that has meaning for an angle value!) each time s increases by unity. Therefore the general theory applies. The introduction of nice variables is of course utterly trivial, being given by

$$\mathbf{z} = \xi, \quad \phi = \theta. \tag{F42}$$

And the reduced system is just (F30) with ξ instead of $\xi^\dagger$. The only point to this otherwise rather farcical merry-go-round is that the further developments of the theory, and especially the proof that the formal series is really an asymptotic representation of the true solution, is secured thereby.

We close with a brief description of the reverse procedure. Had we chosen to develop the mapping theory in full, we could have based the theory of nearly recurrent systems on it by choosing for ξ space a hypersurface in $\mathbf{x}$ space cutting across the loops. The mapping would then be obtained by starting from any point ξ in the hypersurface and following the solution of (B4) around until it first again intersects the hypersurface in a point to be called $\mathbf{T}(\xi)$. Thus we see that the two theories are completely equivalent.

[28] Thus $\mathbf{x}$ space is topologically the product of ξ space with a circle.

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